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New Cambrian and Ordovician Paleomagnetic Poles for the North China Block and Their Paleogeographic Implications

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Multicomponent magnetization has been revealed in several Cambrian and Ordovician sedimentary units from localities in Hebei and Shandong provinces of the North China Block (NCB). A well-defined component with lower unblocking temperature is characterized by northerly declination and moderate to steep inclination with both polarities, and postdates the Late Cretaceous folding. The pole position derived from the overprinted directions is close to both the Cretaceous pole for the NCB as well as the present North Pole. The overprint is not a simple thermal remanent magnetization, but is probably a chemical remanent magnetization imposed at moderately elevated temperature during the major episode of regional deformation in the Late Mesozoic. The higher unblocking temperature characteristic components pass reversal, baked contact, and fold tests, and are internally consistent with rocks of different lithologies from widely distributed localities. Moreover, the data exhibit an overall similarity in the magnetic polarity pattern in the stratigraphic sequence at sampled localities. Hence the high unblocking temperature characteristic component was probably acquired at or near the time of deposition. The Cambrian and Ordovician paleopoles are somewhat different from the other coeval poles for the NCB, which are based on much less data and do not pass as many reliability tests, but all imply that the NCB was in the equatorial zone in the Early Paleozoic.

INTRODUCTION

It is now generally recognized that eastern Eurasia is a composite landmass formed by amalgamation of a number of crustal blocks (Figure 1) [*McElhinny et al.*, 1981; *Klimetz*, 1983; *Opdyke* et al., 1986]. Among these blocks, the North China Block (NCB) is of particular interest in understanding the tectonic history of Asia because of its large area and central position in east Asia. Paleomagnetic data from the NCB would not only provide constraints on the tectonic history of the block itself but also on the overall geologic history and paleogeographic reconstruction of Asia. In recent years, much work has been done in eastern China, and preliminary apparent polar wander paths (APWP's) for the NCB have been compiled [*Lin et al.*, 1985*a*; *Zhou*, 1985; *Zheng et al.*, 1991]. However, Early Paleozoic paleomagnetic data are particularly scarce, and in almost every case the age of magnetization is not demonstrated within standards of acceptable reliability; in addition, no conclusive evidence is presented demonstrating escape from the widespread remagnetization in eastern China [*Kent et al.*, 1987; *Zhao*, 1987]. Because of this and the great sedimentation hiatus during Late Ordovician to Early Carboniferous time on the NCB [*Zh. Zhang et al.*, 1984; *Yang et al.*, 1986], no strong constraints on the locations of the NCB during most of Paleozoic time are available.

We reported two preliminary Cambrian and Ordovician paleomagnetic poles for the NCB from our own preliminary studies at several meetings [*Zhao and Coe*, 1985; *Zhao et al.*, 1989], and gave a tentative interpretation of them in a thesis [*Zhao*, 1987]. In this paper, we present the detailed account of our paleomagnetic results, including those from many new measurements, derived from Latest Precambrian-Early Cambrian to Middle Ordovician shale, limestone, and dolomite from three widely separated localities in Hebei and Shandong provinces. Because our data pass the fold, baked contact, and reversal tests, we believe that the magnetization may be primary, with little or no contamination by unremoved secondary remanent magnetization. We then compare other relevant paleomagnetic data from China and discuss the tectonic and paleogeographic implications.

GEOLOGY OF THE SAMPLED LOCALITIES AND PALEOMAGNETIC SAMPLING

Xishan Area in Hebei Province

Paleomagnetic sampling was restricted to the roadcut exposures near Xiawedian, about 40 km northwest of Beijing (locality 1 of Figure 1), where a spectacular Paleozoic section spanning the Late Precambrian-Cambrian boundary is well exposed. Sampled rocks

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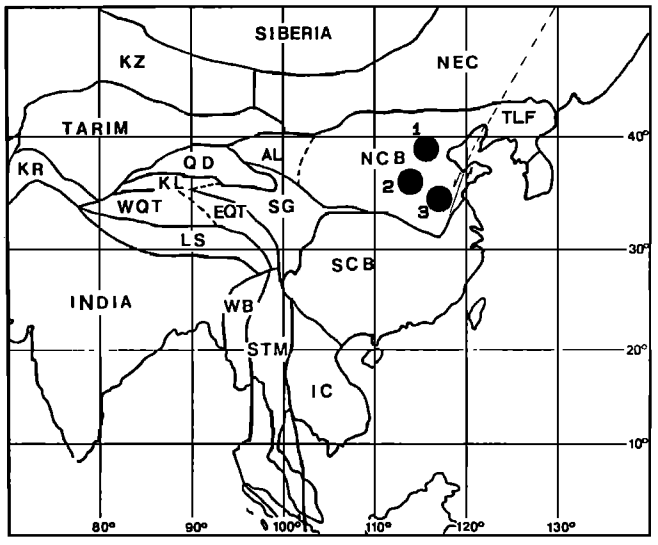


Fig. 1. Sketch map of present location of continental blocks and terranes of China and Eastern Asia [modified from *Lin and Watts*, 1988]. Abbreviations: AL, Alashan/Hexi Corridor Terrane; EQT, East Qiangtang Terrane; IC, Indochina Terrane; KL, Kunlun Terrane; KR, Karakoram Terrane; KZ, Kazakstan Block; LS, Lhasa Terrane; NCB, North China Block; NEC, Northeast China Composite Terrane; QD, Qaidam Terrane; SCB, South China Block; SG, Songpan-Ganzi accretionary complexes; STM, Shan-Thai-Malay Terrane; TLF, Tancheng-Lujiang fault; WB, West Burma Terrane; and WQT, West Qiangtang Terrane. Sampling areas of this study are denoted by numbers: 1, Xishan, Hebei; 2, Jingxin, Hebei; and 3, Wennan and Ji county, Shandong.

consist of purplish-red shale, dark gray limestone, and oolitic limestone in Early Cambrian to Middle Ordovician formations with a thickness of over 800 m. At two sites of the Early Cambrian Mantou Formation, lamprophyre dikes intruded the formation. Similar dikes in the vicinity of Xiawedian, which also intrude Early Cambrian strata, yielded radiometric ages ranging from 136 to 148 Ma [*Chinese Academy of Geological Sciences*, 1973; *Hu et al.*, 1982; *Zh. Zhang et al.*, 1989], corresponding to the Middle Yanshanian (Late Jurassic to Early Cretaceous). Field evidence clearly shows that these dikes intruded after folding. The dike and the contact zone of gray limestone next to it were specially targeted in an attempt to establish the age and the nature of the components of magnetization of the Early Cambrian rocks.

Jingxin Area in Hebei Province

Paleozoic strata of this area were sampled at two separate localities (locality 2 of Figure 1). The first locality is in Nanliangdu village, about 20 km south of Jingxin. This area is about 280 km southwest from Xishan. A continuous section of five formations was sampled with a thickness of more than 400 m. These formations also range in age from Early Cambrian to Early Ordovician and comprise sections of shale, limestone, and dolomite with colors of gray, purple, and red. The bedding strikes of sampled sites are very different from their counterparts in the Xiawedian section (Table 1) and therefore provide the opportunity to perform a regional generalized fold test. The second locality is located in Touquan area, about 15 km northwest of Jingxin. The same Early Paleozoic strata cropped out along roadcuts. The bedding attitudes there are also favorable for performing a fold test (Table 1).

Central Shandong Area in Shandong Province

The Early Paleozoic sedimentary sections in this study area are thick-bedded limestones and dolomites. Sampling sites were distributed in quarries or roadcuts in Wennan, about 30 km northeast of downtown Xingtai (locality 3 of Figure 1). The Cambrian sections consist of five formations of dark gray limestone: oolitic, wurmkalk limestones occasionally interbedded with calcareous shale. The Early Ordovician stratigraphy is dominated by fine- to medium-grained dolomite and limestone, mostly gray in color. These formations overlie the Late Cambrian limestone Fengshan Formation with a gradational contact zone that seems to record continuous deposition across the Cambrian-Ordovician boundary. The Middle Ordovician limestone of the Majiagou Formation overlies these formations. About 100 km to the southeast in Ji county, which is located within the Yishu fracture zone (the main portion of the Tancheng-Lujiang fault zone), a Latest Precambrian to Early Cambrian argillaceous shale and marl unit was also sampled. This unit contains typical Late Precambrian-Cambrian fossils (e.g., *Collenia sinensis* and *Collenia cylinderica*) and represents the uppermost section of the Tumen Formation which spans the Late Precambrian-Cambrian boundary [*Chinese Academy of Geological Sciences*, 1973; *All-China Stratigraphic Committee*, 1980].

Dolomitization

Most of the dolomites in Hebei province have been interpreted as being of early origin because they have porosities and petrologic textures similar to the limestones with which they are intercalated [*Lu et al.*, 1976; *Lai et al.*, 1982]. These dolomites are reported to contain *Nautiloids*, and laterally extend over a wide area. Secondary dolomitization may also exist in the sampling areas, but we have not seen significant signs of recrystallization in sampled beds. In Shandong province, fine-grained dolomites overlie the Late Cambrian limestone units with a gradational contact, and are overlain abruptly by the Middle Ordovician Majiagou limestone. The sharpness of the contact would suggest that dolomitization probably occurred early, most likely in pre-Majiagou (Middle Ordovician) time, because late dolomitization would also affect the Majiagou limestone and would not exhibit such nice stratigraphic control.

Fold and Sampling

In all three sampled areas, although strict local fold tests were not available for all components of magnetization, the structural variations at each area allowed local tilt tests to be applied on these components. In addition, the three areas provide in combination an excellent regional fold test (see Table 1). The age of the folding of these Paleozoic strata in all three sampled areas occurred first in Triassic (Indosinian orogeny) and again during Jura-Cretaceous time (Yanshanian orogeny), as suggested by regional unconformities and other geologic data [*Yang et al.*, 1986; *Nie*, 1985].

Oriented samples were obtained in the field with a portable gasoline-powered drill during three field seasons. Sampling sites, each represented by at least six or more independently oriented samples, were distributed over these three general sampling locations, mostly from fresh roadcuts. The samples were oriented

TABLE 1. Paleomagnetic Sampling Details of Late Precambrian-Cambrian and Ordovician Sites in Four Areas of Hebei and Shandong Provinces

Site ID	Age	Rock Type (Formation) and Locality	SLat(N)	SLong(E)	N	Strike	Dip	
		Hebei Province						
Xishan Area								
NCB.001	CB1	Red shale (Mantou), Xiawedian	40.0	116.0	15	78	35	SE
NCB.002	CB1	Red shale/limestone (Mantou), Xiawedian	40.0	116.0	13	92	32	SE
NCB.003	CB2	Oolitic limestone (Zhangxia), Xiawedian	40.0	116.0	10	74	50	SE
NCB.004	CB3	Limestone (Changshan), Nylon quarry	40.0	116.0	11	100	45	SW
NCB.005	CB3	Limestone (Changshan), Nylon quarry	40.0	116.0	11	85	45	SE
NCB.006	O1-2	Limestone (Yeli-Majiagou), Dingjiatan	40.0	116.0	72	88	42	SE
NCB.007	O1-2	Limestone (Yeli-Majiagou), Dingjiatan	40.0	116.0	35	310	15	NE
Jingxin Area								
NCB.025	CB1	Red shale (Mantou), Nanliangdu	38.0	114.1	5	176	28	SW
NCB.026	CB1	Shale/limestone (Mantou), Touquan	38.0	114.2	18	240	65	NW
NCB.027	CB1	Dolomite (Mantou), Nanliangdu	38.0	114.1	6	176	31	SW
NCB.028	CB1	Mudy dolomite (Mantou), Nanliangdu	38.0	114.1	6	176	31	SW
NCB.029	CB2	Dolomite (Maozhuang), Nanliangdu	38.0	114.1	6	174	26	SW
NCB.030	CB2	Limestone (Xuzhuang), Nanliangdu	38.0	114.1	6	183	30	NW
NCB.031	CB2	Oolitic limestone (Zhangxia), Touquan	38.0	114.2	19	248	46	NW
NCB.032	CB2	Limestone (Zhangxia), Nanliangdu	38.0	114.1	6	202	20	NW
NCB.033	CB2	Limestone (Zhangxia), Nanliangdu	38.0	114.1	6	183	25	NW
NCB.034	CB3	Limestone (Gushan), Nanliangdu	38.0	114.1	6	200	20	NW
NCB.035	CB3	Limestone (Fengshan), Nanliangdu	38.0	114.1	6	195	20	NW
NCB.036	CB3	Limestone (Fengshan), Xiaan	38.0	114.2	12	236	36	NW
NCB.037	CB3	Limestone (Changshan), Touquan	38.0	114.2	11	232	36	NW
NCB.038	O1	Limestone (Liangjiashan), Xiaan	38.0	114.2	13	223	35	NW
NCB.039	O1	Limestone/shale (Liangjiashan), Xiaan	38.0	114.2	10	232	34	NW
NCB.040	O1	Limestone (Yeli), Xiaan	38.0	114.2	9	236	32	NW
NCB.041	O1-2	Limestone (Yeli-Majiagou), Shangan	38.0	114.2	14	74	23	SE
NCB.042	O2	Limestone (Majiagou), Shangan	38.0	114.2	23	215	22	NE
NCB.043	O2	Limestone (Majiagou), Shangan	38.0	114.2	8	56	20	SE
		Shandong Province						
Ji County								
NCB.067	Prc	Marl (Tumen), Ji county	35.6	118.9	10	350	70	NE
Wennan Area								
NCB.069	CB1	Red sandstone (Maozhuang), Wennan	35.9	117.9	9	312	18	NE
NCB.073	CB3	Limestone (Changshan), Wennan	35.9	117.9	9	295	16	NE
NCB.074	CB3	Limestone (Fengshan), Wennan	35.9	117.9	13	310	19	NE
NCB.075	O1	Dolomite (Zifangzhuang), Wennan	35.9	117.9	7	300	20	NE
NCB.078	O1	Limestone (Majiagou), Wennan	35.9	117.9	13	304	20	NE
NCB.079	O2	Limestone (Gezhuang), Wennan	35.9	117.9	12	307	27	NE
NCB.080	O2	Limestone (Badou), Wennan	35.9	117.9	14	313	19	NE

Site ID, site identification; Age, approximate age of rocks studied, based mainly on paleontological and geological evidence. Prc, Precambrian, CB, Cambrian, O, Ordovician; 1, Early, 2, Middle, 3, Late. Rock type (Formation) and Locality, rock type, name of rock formation, name of sampling locality; SLat(N), site latitude; SLong(E), site longitude; N, number of samples sampled; strike, bedding strike azimuth; dip, bedding dip, the direction of dipping is 90° clockwise from indicated bedding strike azimuth.

using a Brunton compass that can be mounted on an orienting device. Solar orientations were also taken at each site. Although the sun's azimuthal readings could not be taken for all samples (because of the weather), solar and magnetic orientations were usually consistent within 1°. Bedding attitudes were measured and recorded directly at each sampling site.

LABORATORY PROCEDURES AND DATA ANALYSIS

Samples were measured in the UCSC paleomagnetic laboratory. Measurements were done with a Schonstedt spinner magnetometer and a 2G cryogenic magnetometer housed in a field-free room. Both thermal and alternating field (AF) demagnetization was performed with Schonstedt equipment to evaluate the directional stability and coercivity/unblocking temperature spectra of each site. In most cases, thermal demagnetization was more successful in isolating the various components of magnetization. On the basis of more detailed thermal demagnetization experiments on pilot samples from each site, the remaining samples were thermally demagnetized at a minimum of nine temperature intervals. The temperature intervals ranged from as large as 300°C at initial thermal demagnetization temperature to as small as 5°C above 650°C. For some limestones that may incur thermochemical alteration at higher temperature demagnetization, a hybrid

demagnetization, which comprises a particular strategy with partial thermal treatment followed by AF demagnetization, turned out to be very efficient in cleaning.

For each sample, a vector diagram was used to identify the multiple components of magnetization that were present. When the coercivity or unblocking temperature spectra of the different components were sufficiently distinct, identification of each component was done by fitting least-squares lines to segments of the vector demagnetization plots [*Zijderveld*, 1967; *Kirschvink*, 1980] that were linear in three-dimensional space. In a few cases where curvature of the vector demagnetization trajectories that did not reach the origin gave rise to a suspicion that complete separation of components was not achieved, a great circle technique of demagnetization planes was used to establish the characteristic direction [*McFadden and McElhinny*, 1988]. Statistics on the distributions of paleomagnetic directions were calculated by using standard *Fisher* [1953] statistics and the new method of combined data from the remagnetization plane and least- squares- best- fit line, as described by *McFadden and McElhinny* [1988]. The overall paleomagnetic poles were calculated from virtual geomagnetic poles (VGPs) derived from paleomagnetic directions, as VGPs often are more nearly Fisherian distributed than are directions [*Cox*, 1970; *Harrison*, 1980].

PALEOMAGNETIC RESULTS OF THE XISHAN AREA IN HEBEI PROVINCE

Cambrian Sites (NCB.001-005 in Table 1)

A common magnetic property of the Cambrian rocks from this area is that the natural remanent magnetization (NRM) intensities are typically of the order 1 mA/m for the Early Cambrian red shale of the Mantou formation, and of the order 10^{-1} mA/m for the Mid-Late Cambrian gray limestone of the Zhangxia and Changshan formations. The site mean NRM directions fall in the northwest quadrant of the stereonet, quite different from the present-day field direction. After removal of a stray viscous remanent magnetization (VRM) component in the initial stages of demagnetization, almost all samples from the six sites exhibit two distinct components of magnetization. One is directed to the north-northwest with a positive inclination in geographic (in-situ) coordinates at low to intermediate thermal demagnetization steps (300 to 600°C). This component is termed component *A*. A second component is resolved at temperatures above 625°C that is directed toward the western quadrant with a negative inclination in geographic coordinates (Figure 2). However, two specimens in site NCB.001 appear to show both polarities of the component; their demagnetization paths finally reached the southeasterly quadrant and positive inclinations during thermal demagnetization to 700°C. The second component is hereafter identified as the characteristic component (ChRM).

There is a baked contact test available in the locality where site NCB.001 and 002 were sampled. Thermal and AF demagnetization on five limestone samples, about 5 m away from a dike, clearly showed the effect of baking. The dike samples yielded only one component of magnetization (Figure 3) with D = 343.9°, I = 60.9°, N = 4 specimens, α_{95} = 13.3° , and k = 48.5. The baked limestone samples gave the same in-situ direction (not statistically distinguishable at 95% confidence level) as the dike (see Table 2), and this direction agrees very well with that of the

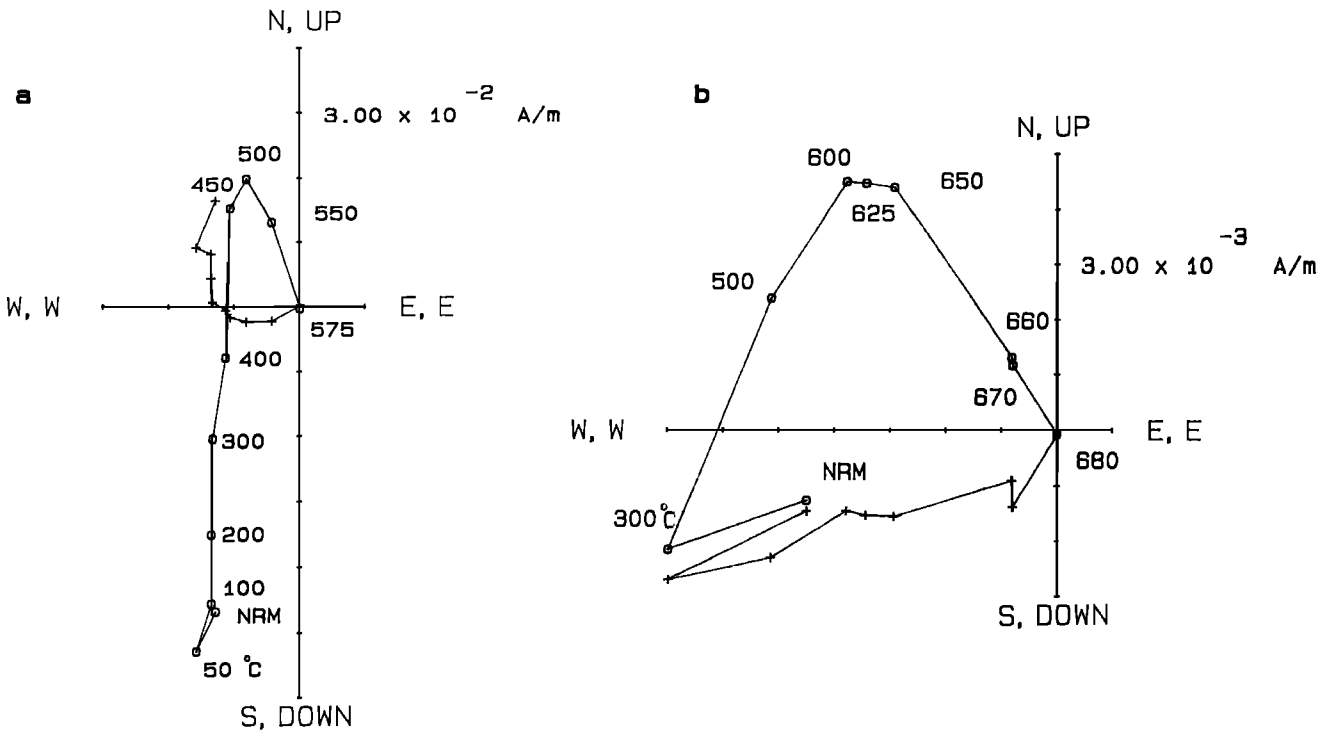


Fig. 2. Orthogonal vector plots of representative samples from Cambrian sites in the Xishan area: (*a*) sample from site NCB.003 showing the *A* and ChRM components that probably reside in magnetite; (*b*) thermal demagnetization of one red shale sample from site NCB.002 reveals that the ChRM component is probably carried by hematite. Directions are plotted in geographic coordinates. Crosses represent vector endpoints projected onto the horizontal plane; open circles, endpoints projected onto the vertical plane.

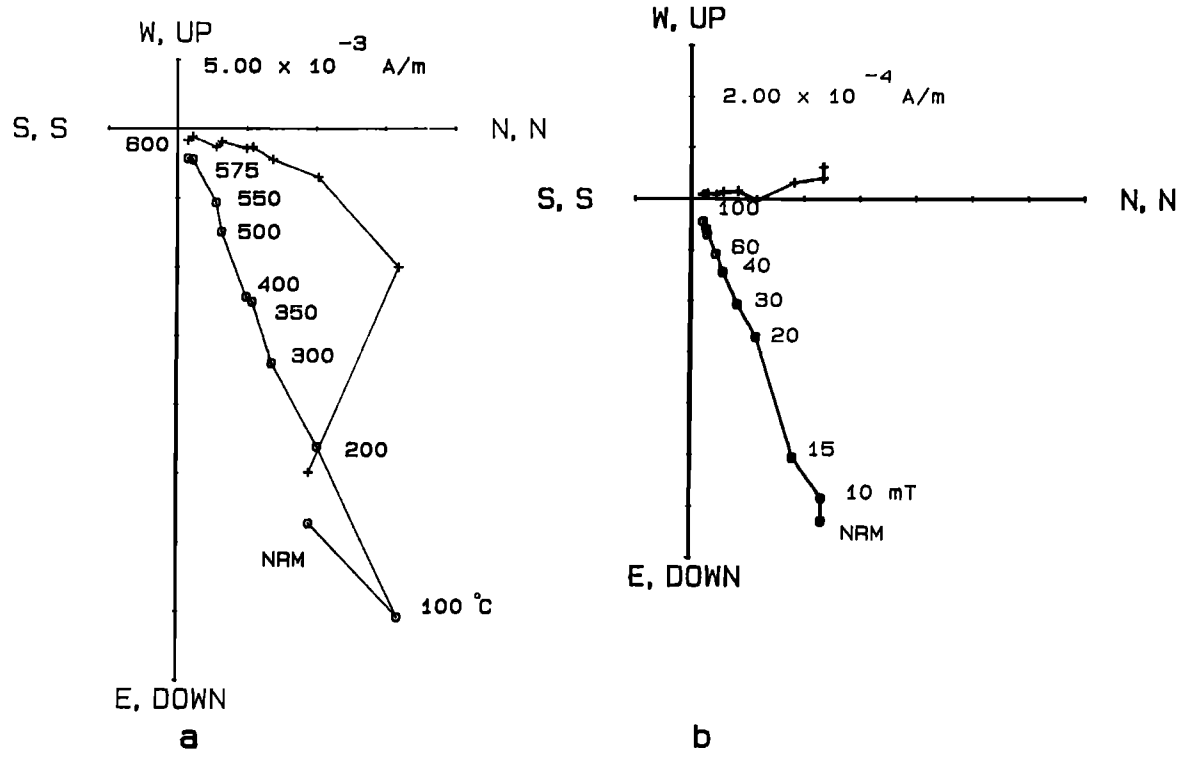


Fig. 3. Demagnetization behavior in the baked contact zone in sites NCB.001 and NCB.002. (*a*) Typical vector plot of thermal demagnetization of dike samples; (*b*) vector plot of AF demagnetization of one baked limestone. Directions are plotted in geographic coordinates. Crosses and open circles same as Figure 2.

Cretaceous for this area [*Cheng et al.*, 1988]. However, this direction is significantly different from the ChRM component isolated from the red shale samples that are further from the dike (Table 2). The baked contact test results therefore provide indirect evidence that the ChRM component was unheated by the dike, and was probably imposed earlier than the intrusion of the dike in Jura-Cretaceous time.

TABLE 2. Baked Contact Test in the Xishan Area, Hebei Province

In Situ Direction	*N*	Direction					Pole Position			
		D	I	*a*95	*k*	*R*	Long.	Lat.	*A*95	*K*
Dike	4 specimens	343.9	60.9	13.3	48.5	3.93814	42.7	77.7	19.1	24.1
Baked LM	5 specimens	2.9	72.1	9.5	65.4	4.93881	121.1	72.0	16.0	23.9
DYKLM	9 specimens	352.3	67.4	7.9	43.8	8.81729	95.1	77.6	12.0	19.2
A	4 sites	0.8	71.4	4.8	361.5	3.99170	117.2	73.8	8.3	124.9
ChRM-CB	2 sites	262.6	-56.2	-	345.7	1.99711	359.7	-27.7	-	311.3
ChRM-O	4 samples	244.4	-65.2	12.9	51.4	3.94116	354.3	-45.1	18.8	24.8

F-Test on Directions	F-table (95%)	F-statistics	Y/N
Dike vs. Baked LM	F(2,14) >3.74	3.40	Y
DYKLM vs. *A*	F(2,22) >3.42	0.60	Y
DYKLM vs. ChRM-CB	F(2,18) >3.52	165.59	N
DYKLM vs. ChRM-O	F(2,22) >3.42	254.50	N

Abbreviation: Dike, mean of dike specimens; Baked-LM, mean of baked limestones; DYKLM, mean of dike and baked limestones; *A*, mean of the *A* component; ChRM-CB/ChRM-O, mean of the ChRM component from Cambrian/Ordovician sites; *N*, number of specimens/samples/sites; D/I, delination/inclination; Long./Lat., east longitude and north latitude of virtual geomagnetic pole (VGP); a95/A95, radius of circle of 95% confidence about the direction/VGP in degrees; *k*/*K*, Fisher precision parameter for direction/VGP; *R*, length of resultant vector; Y/N, compared directions are statistically indistinguishable (Y)/distinguishable (N) at 95% confidence level.

Ordovician Sites (NCB.006-007 of Table 1)

The Ordovician limestone samples were collected from both limbs of a gently plunging fold (sites NCB.006 and NCB.007). The fold axis plunges to the north only about eight degrees, which produces less than three degrees of correction for the bedding attitudes of the two limbs sampled [*Norman*, 1960]. The NRM intensities of these sampled limestones were typically about 5 mA/m, about 5 times stronger than the limestone samples of Late Cambrian age. Thermal and AF demagnetization methods were both successful in isolating the *A* component, which in this case decays univectorially toward the origin of the vector plots (Figure 4). The fact that AF demagnetization on these limestone samples was so effective in isolating the *A* component indicates that the *A* component probably resides in magnetite.

Most of the samples in this locality seem to have lost their memory of the ChRM component. Out of 28 demagnetized samples, only four samples recorded the ChRM direction, which is characterized by westerly declination and negative inclination (Figure 5).

PALEOMAGNETIC RESULTS OF THE JINGXIN AREA IN HEBEI PROVINCE

Cambrian Sites (NCB.025-034 of Table 1)

The paleomagnetic components of the Cambrian formations from both the Nanliangdu and Touquan localities in many ways resemble each other and their counterparts in the Xishan area. Specifically, the same low-(*A*) and high-(ChRM) temperature components are usually present. However, the demagnetization behavior is quite different, as shown by the generally more complicated demagnetization paths (Figure 6). Three major differences in the paleomagnetic signals were observed. First, the *A* component is usually removed over progressive demagnetization temperatures ranging up to 350°C, considerably lower than those for the sections in the Xishan area. A reversed *A* component is also present with directions antiparallel to the majority of the normal *A* directions in the same section, and sometimes in the same sample (Figure 6). On the other hand, the ChRM component is commonly revealed above 575°C thermal demagnetization. Second, upon progressive thermal demagnetization, samples from many sites are often found not to have the *A* component (Table 3). This is especially common for sites in the Nanliangdu area. Third, the percentage of reversed-polarity ChRM component in the Jinxin area is much higher than that in the Xishan area, and the distribution of normal and reversed polarities appears to be arranged in stratigraphic sequence. The occurrence of both polarity states of the magnetization increases the chance that the ChRM is an ancient magnetization.

Ordovician Sites (NCB.038-043 of Table 1)

The Ordovician rock units from this area are overprinted strongly by the *A* component, so that only one site yielded the high temperature ChRM component (site NCB.038, see Table 4). The *A* component is easily resolved from these Ordovician limestones by both AF and thermal demagnetization. The demagnetization paths commonly tend toward the origin of the vector plots, indicating that the *A* component is the only component present.

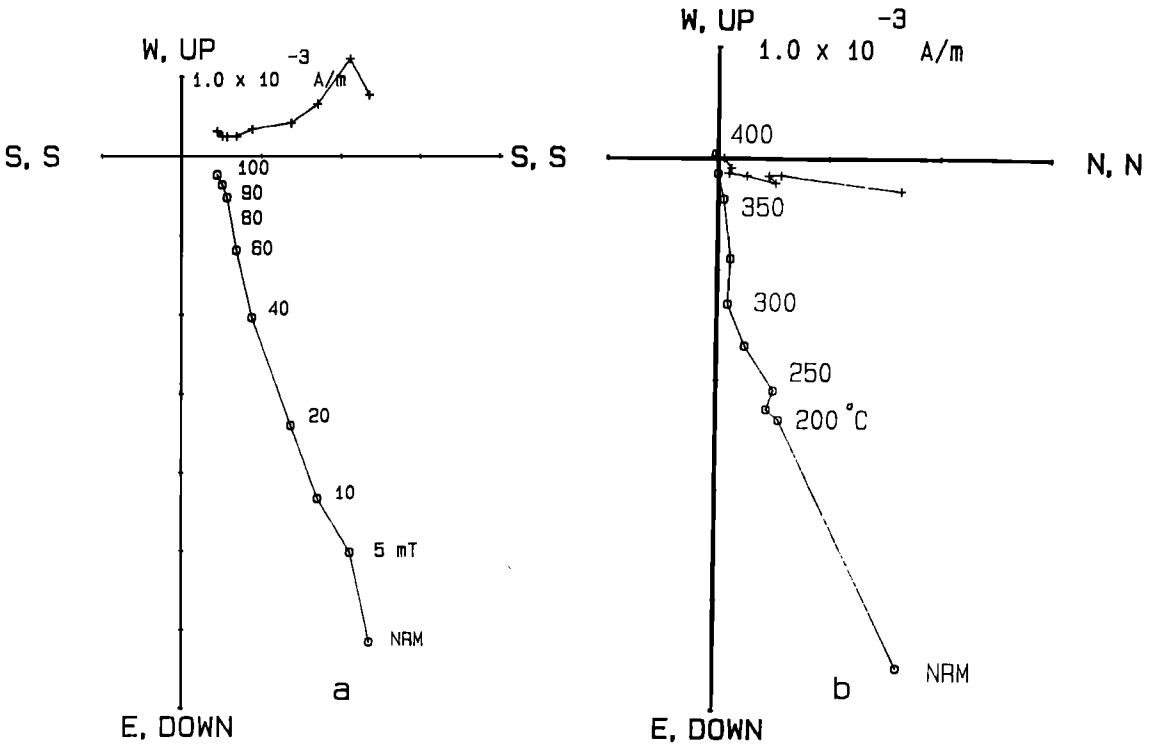


Fig. 4. Vector plot of in situ directions of Ordovician limestone site NCB.006 from the Xishan area (*a*) during AF demagnetization and (*b*) during thermal demagnetization. Directions are plotted in geographic coordinates. Crosses and open circles same as Figure 2.

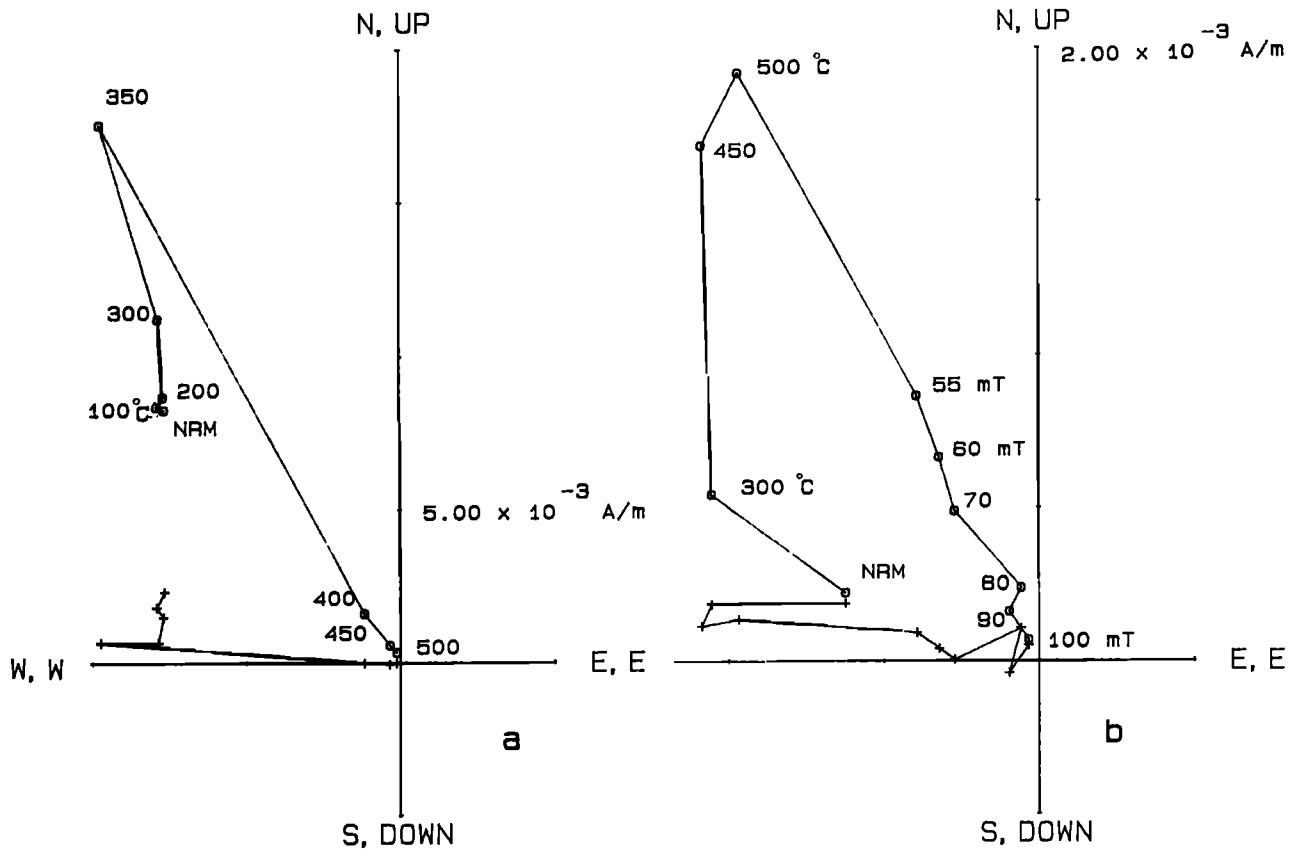


Fig. 5. (*a*) Thermal demagnetization of sample 82X0722A from site NCB.006, displaying the ChRM component. (*b*) Representative vector plots of hybrid demagnetization of Middle Cambrian limestone, thermal demagnetization to 500° first, followed by AF demagnetization. Directions are plotted in geographic coordinates. Crosses and open circles same as Figure 2.

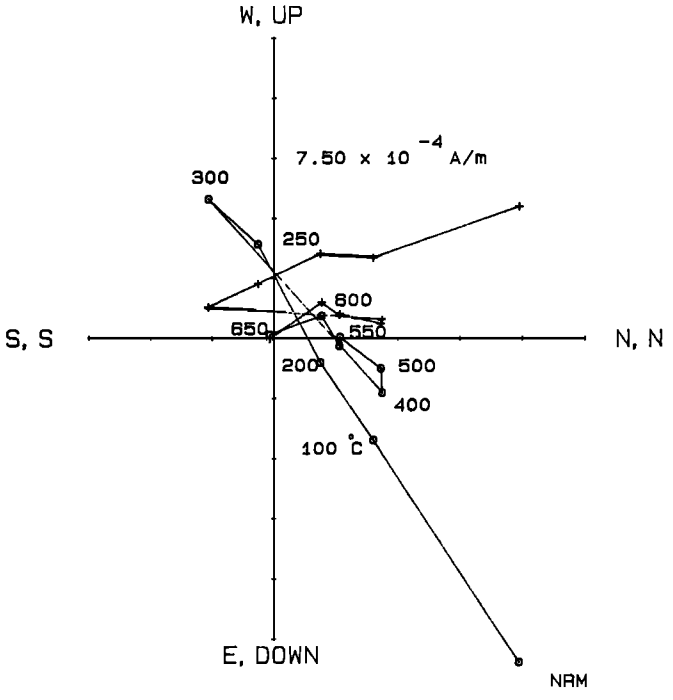


Fig. 6. Vector plot of magnetic directions for Early Cambrian limestone sample 83X0959A from the Jingxin area following thermal demagnetization, revealing at least three components (normal and reversed *A*, and ChRM) in this sample. Directions are plotted in geographic coordinates. Crosses and open circles same as Figure 2.

The *A* component from this locality fails the standard fold test; however, the site mean directions in geographic and stratigraphic coordinates cross after the bedding correction is made. This suggests that the magnetic directions from opposite limbs pass near a common point during correction for bedding tilt. The results of an incremental fold test support this view and suggest that the *A* component occurred during folding (Figure 7*a*).

PALEOMAGNETIC RESULTS OF THE CENTRAL SHANDONG AREA IN SHANDONG PROVINCE

Cambro-Ordovician Sites (NCB.069-080 of Table 1)

Like the situation in Hebei province, the *A* component is widely present in the Cambro-Ordovician sites sampled in Wennan, central Shandong, with the majority of the directions being of normal polarity. An incremental fold test on the *A* component from both limbs of a local fold once again suggests that the component was probably imposed during folding (Figure 7*b*). Unfortunately, the high temperature ChRM component could only be resolved from one Late Cambrian and two Ordovician sites (see Table 4). The NRM intensities of these three Cambro-Ordovician limestone and dolomite sites are more variable than that of sampling areas in Hebei, ranging from 10^{-2} to 1 mA/m, but most samples display relatively simple demagnetization behavior, characterized by the removal of a large portion of the *A*

component over a temperature range from 200 to 575°C and isolation of the ChRM component with both polarities (Figure 8). It is interesting that the ChRM isolated from these sites is dominantly "reversed-polarity" (south and downward), which is in good agreement with the polarity distribution in the Jingxin area in Hebei province.

In samples from sites NCB.074, 078, and 079, the final ChRM component is certainly present, as shown either by one or two

TABLE 3. Overprint (*A*) Component Site Mean

Site	GDec	GInc	GLong	GLat	SDec	SInc	SLong	SLat
Xishan								
Dike Site	1.5	68.3	120.7	78.5	1.5	68.3	120.7	78.5
NCB.003	346.3	74.7	99.0	67.1	162.9	55.3	130.1	-12.6
NCB.004	356.5	69.8	107.3	76.1	173.8	65.2	120.2	-2.6
NCB.005	3.6	70.2	124.5	75.6	168.3	64.4	124.1	-3.1
NCB.006	13.5	70.1	144.3	73.2	164.7	66.7	125.9	0.3
NCB.007	3.1	72.1	121.7	72.7	19.1	58.8	201.6	75.3
Jingxin								
NCB.043	8.0	54.8	224.4	83.1	38.3	66.1	171.0	60.6
NCB.042	20.3	60.4	185.7	74.1	352.2	49.4	336.3	79.9
NCB.041	1.1	48.4	287.8	81.3	17.8	69.4	162.6	70.9
NCB.040	31.9	65.3	171.9	65.0	357.3	42.8	304.5	76.7
NCB.039	28.7	69.1	158.5	65.3	349.1	44.1	334.2	74.8
NCB.038	25.9	63.9	174.4	69.4	347.1	41.4	335.7	72.0
NCB.037	3.3	67.6	123.8	77.3	340.0	35.4	343.0	64.6
NCB.036	1.5	70.7	117.1	73.0	340.0	37.3	345.1	65.5
NCB.031	8.3	62.0	161.8	81.8	352.5	18.8	309.5	60.9
NCB.026	351.4	59.3	44.3	83.0	340.8	-3.9	322.5	46.3
Ji County								
NCB.067	293.7	39.2	35.0	31.5	31.0	55.3	199.2	64.9
Wennan Area								
NCB.078	2.6	59.8	140.3	84.8	13.4	41.7	250.0	73.4
NCB.079	4.6	68.5	128.1	73.8	21.3	43.5	231.9	68.9
NCB.080	348.1	69.2	95.2	71.2	12.7	54.8	207.3	79.7
NCB.069	349.5	66.0	89.2	75.4	10.1	52.4	224.2	81.2
NCB.075	4.0	70.3	125.2	71.3	16.3	51.4	219.1	76.0
NCB.074	358.7	67.8	114.7	75.1	16.4	51.5	218.6	75.9
NCB.073	351.1	68.3	98.5	73.2	4.5	54.0	227.0	86.1

Average

Locality	N	C	D	I	a_{95}	k	Long.	Lat.	A_{95}	K
Xishan	6	*G*	*1.3*	*71.0*	*3.0*	*510.8*	*118.5*	*74.4*	*5.0*	*181.9*
		S	156.7	79.6	24.3	8.6	128.2	21.2	42.7	3.4
Jingxin	10	G	11.1	62.7	5.5	77.3	162.3	79.0	7.5	42.2
		15%	*6.0*	*59.9*	*5.0*	*95.6*	*168.9*	*84.1*	*6.6*	*54.6*
		S	351.1	41.2	14.7	11.7	321.1	76.7	13.4	13.9
Hebei	16	G	8.2	65.9	4.0	86.6	142.1	78.2	5.3	49.0
		S	354.2	62.0	18.8	4.8	115.4	78.5	26.1	3.0
Wennan	7	G	357.1	67.3	3.3	331.6	110.3	75.5	4.8	159.0
		S	13.8	50.0	4.6	171.4	227.2	77.6	4.8	161.5
Shandong	8	G	344.3	66.2	11.1	26.0	87.4	73.7	14.2	16.2
		S	*15.7*	*50.8*	*4.8*	*131.9*	*221.9*	*76.3*	*5.5*	*102.9*
Grand average										
	24	G	0.4	66.4	4.5	44.6	120.8	78.1	5.9	26.2
		S	4.6	58.4	12.2	6.9	162.6	82.4	16.3	4.3
Average with italicized means										
OVP	3		9.2	60.7		57.2	166.4	81.9		48.7

Late Cretaceous Paleomagnetic Poles for North and South China (NCB and SCB) and Eurasia (EUR)

	N	*R*	Long	Lat	A95
NCB*	3	2.99	242.3	83.5	3.5
SCB#	5	4.94	213.7	82.9	9.3
EUR1+	8	7.89	184.0	81.0	7.0
EUR2$	4	3.99	192.2	76.7	4.0

F-Test on Poles	F-table (95%)	F-statistics	Y/N
OVP vs. NCB	F(2, 8) > 4.46	1.43	Y
OVP vs. SCB	F(2, 12) > 3.89	0.63	Y
OVP vs. EUR1	F(2, 18) > 3.52	0.15	Y
OVP vs. EUR2	F(2, 10) > 4.10	1.72	Y

GDec/GInc, declination and inclination in geographic coordinates; GLong/GLat, east longitude and north latitude of VGP in geographic coordinates; SDec/SInc, declination and inclination in stratigraphic coordinates; SLong/SLat, east longitude and north latitude of VGP in stratigraphic coordinates; *N*, number of sites exhibiting the *A* component; C, coordinate system for component *A*: (G) Geographic, (S) Stratigraphic, (15%) *k*-maximum mean direction at 15% unfolding; Other abbreviations same as Table 2;

*, Average poles for Outer Mongolia [*Pruner*, 1987], Inner Mongolia [*Zhao et al.*, 1990], and North China [*Xiong et al.*, 1987; *Tan*, 1988; *Cheng et al.*, 1988]. See *Zhao et al.* [1990] for detailed discussion.

#, Average poles given by *Kent et al.* [1986], *Zhuang et al.* [1988], *Enkin et al.* [1991], and *Chan* [1991].

+, Average eight transferred poles from Africa, see *Westphal et al.* [1986].

$, The 70-100 Ma poles of the *Besse and Courtillot* [1991] APWP for Eurasia.

TABLE 4. Characteristic (ChRM) Site Mean Directions

							Direction				Pole Position			
Site	Age	P	n/No	R/N	ALY	C	D	I	a95	*k*	Long	Lat	*A*95	*K*
							Hebei Province							
Xishan Area														
NCB.001	CB1	M	13/14	2/1	LSF	G	267.1	-55.8	7.6	30.9	357.2	-24.5	6.8	37.9
						S	306.3	-31.1	7.6	30.9	348.8	13.9	6.8	37.9
NCB.002	CB1	R	11/11	11/0	LSF	G	261.7	-59.2	5.5	69.2	356.6	-30.5	7.1	42.5
						S	308.1	-30.0	5.5	69.5	347.6	15.9	4.2	117.7
NCB.003	CB2	R	8/13	8/0	LSF	G	285.1	-65.1	9.1	38.3	340.5	-21.0	13.1	18.9
						S	320.6	-24.6	9.1	38.3	339.3	25.3	5.8	92.9
NCB.004	CB3	R	6/6	6/0	LSF	G	302.3	-60.8	11.6	34.3	335.4	-7.9	16.4	17.7
						S	329.3	-23.8	11.6	34.3	330.8	30.0	9.2	53.4
NCB.005	CB3	R	10/10	10/0	LSF	G	272.0	-71.4	6.9	49.8	336.1	-31.4	11.9	17.5
						S	331.7	-40.5	7.0	49.0	324.1	21.2	7.2	46.2
NCB.006	O1-2	R	4/14	4/0	LSF	G	244.4	-65.2	12.9	51.4	354.3	-45.1	18.8	24.8
						S	319.4	-51.9	12.9	51.4	328.8	7.9	12.4	55.5
Jingxin Area														
NCB.025	CB1	R	4/5	4/0	LSF	G	323.1	-25.5	14.0	43.9	335.3	27.8	10.6	76.0
						S	339.1	-37.6	14.0	43.9	315.8	27.4	13.2	49.2
NCB.026	CB1	R	5/10	5/0	LSF	G	0.2	40.4	12.0	41.8	293.6	75.4	12.8	36.9
						S	354.0	-19.0	12.0	41.7	302.1	41.7	10.6	53.4
NCB.027	CB1	M	3/6	1/2	LSF	G	321.7	-7.0	26.1	23.4	343.3	35.4	18.2	47.1
						S	329.0	-23.1	26.1	23.4	330.2	32.1	23.0	29.7
NCB.028	CB1	N	2/6	0/2	LSF	G	116.1	11.4	-	89.6	182.7	-16.5	-	110.2
						S	124.2	37.4	-	90.0	166.0	-11.2	-	69.1
NCB.029	CB2	N	4/6	0/4	LSF	G	106.5	7.0	11.2	68.4	191.0	-10.7	8.9	107.1
						S	110.3	30.8	11.2	68.5	178.5	-4.8	9.6	92.4
NCB.030	CB2	R	5/5	5/0	LSF	G	322.2	-27.6	12.9	36.0	335.4	26.1	9.7	63.1
						S	341.2	-43.7	12.9	36.1	312.1	23.3	12.4	38.9
NCB.034	CB3	R	2/6	2/0	LSF	G	334.5	-31.8	-	20.8	322.4	29.8	-	-
						S	346.8	-44.0	-	20.8	307.1	24.6	-	-
							Mean Cambrian Sites in the Jingxin Area							
Normal	CB	N	2/7			G	111.3	9.2	-	120.4	186.9	-13.6	-	136.9
						S	116.9	34.2	-	74.6	172.3	-8.1	-	69.1
Reversed	CB	R	5/7			G	331.9	-11.9	33.6	6.2	330.6	39.6	22.6	12.4
						S	342.8	-33.7	13.9	31.3	314.9	30.3	10.9	50.6
Mixed	CB	M	7/7			G	138.8	11.6	26.4	6.2	163.7	-32.7	21.3	9.0
						S	149.4	35.8	16.7	14.0	145.8	-25.1	17.7	12.6
NCB.038	O1	N	3/12	0/3	LSF	G	177.0	15.2	10.9	128.3	118.4	-44.2	11.2	123.4
						S	191.0	37.8	10.9	129.3	102.5	-29.9	11.6	114.5

Site	Age	P	n/No	R/N	ALY	C	Direction D	Direction I	Direction a95	Direction k	Pole Position Long	Pole Position Lat	Pole Position A95	Pole Position K
							Mean Cambrian Sites in the Hebei Province							
Normal	CB	N	2/2			G	111.3	9.2	-	120.4	186.9	-13.6	-	136.9
						S	117.0	34.3	-	74.9	172.3	-8.1	-	69.1
Reversed	CB	R	10/10			G	313.6	-42.5	26.8	4.2	338.5	7.2	25.2	4.6
						S	329.8	-32.5	9.7	25.9	327.2	26.5	9.9	24.8
Mixed	CB	M	12/12			G	128.0	36.3	23.4	4.4	163.9	-8.6	21.3	5.1
						S	144.4	33.4	9.9	20.1	151.8	-23.7	10.4	18.3
							Mean Ordovician Sites in Hebei Province							
Mixed	O	M	2/2			G	151.3	52.7	-	1.9	146.0	0.5	-	1.3
						S	168.6	47.8	-	9.1	127.2	-20.4	-	5.6
							Shandong Province							
Ji County Area														
NCB.067	Prc-CB1	N	4/7	0/4	LSF	G	227.7	37.6	11.4	66.5	71.5	-16.9	10.2	81.8
						S	135.5	54.7	11.4	66.1	153.3	-7.7	16.1	33.6
			4*/7	0/4	GCA	G	232.9	50.1	11.6	232.4	74.3	-6.9	14.2	154.6
						S	113.0	41.5	6.7	692.2	175.4	-3.1	5.5	1032.0
			8*/7	0/8	CMB	G	228.1	40.2	6.9	70.9	72.4	-15.4	6.5	80.2
Wennan Area						S	130.6	50.1	8.6	46.1	158.9	-8.6	10.6	30.6
NCB.073	CB3	M	8/9	1/7	LSF	G	180.9	20.3	6.4	143.3	116.7	-43.6	5.1	228.3
						S	177.3	34.8	6.4	142.2	121.1	-34.9	5.9	170.2
NCB.075	O1	N	4/7	0/4	LSF	G	181.0	18.9	5.5	277.9	116.6	-44.4	4.3	466.0
						S	175.4	36.0	5.5	279.3	123.1	-34.0	5.1	329.3
NCB.080	O2	R	4/12	4/0	LSF	G	359.0	-7.7	11.6	63.4	299.5	47.6	10.2	82.0
						S	355.4	-21.1	11.6	63.5	303.9	40.4	11.2	68.0
							Grand Average							
Mean Cambrian sites														
Normal		N	4/14			G					145.7	-30.9	77.1	2.4
						S					157.8	-15.3	31.5	9.5
Reversed		R	10/14			G					338.5	7.2	25.2	4.6
						S					327.2	26.5	9.9	24.8
Mixed		M	14/14			G					155.6	-13.2	23.5	3.8
						S					149.9	-23.5	10.1	16.3
Mean Ordivician sites														
Normal		N	3/5			G					116.8	-41.9	6.5	358.4
						S					117.5	-31.3	18.2	47.1
Reversed		R	2/5			G					328.4	2.9	-	1.2
						S					318.3	26.0	-	7.8
Mixed		M	5/5			G					127.6	-31.3	46.5	3.7
						S					125.8	-29.1	18.6	17.9

Overall Averaging by Site

	N	C	Long.	Lat.	A95	K
Cambrian (Prc-CB1-CB2)	10	G	341.2	14.9	30.5	3.5
		S	335.2	21.2	12.4	16.2
Ordovician (CB3-O1-O2)	8	G	318.7	18.5	32.8	3.8
		S	310.9	28.8	12.3	21.1

F-test on Cambrian paleopoles for the NCB

Studies	F-table (95%)	F-statistics	Y/N
This study vs. *Lin and Fuller* [1990]	F(2, 30) > ~3.3	15.11	N
This study vs. *Gao et al.* [1983]+	F(2, 26) > 3.35	2.82	Y
This study vs. *Wu* [1988]	F(2, 64) > ~3.0	68.88	N

Age, see Table 1 for explanation of abbreviation; P, polarity; R, reversed; N, normal; M, both N+R; n/No, number of samples, sites, areas, or regions exhibiting the ChRM component and used in statistical analysis/total number of the samples, sites, areas, or regions; R/N, polarity ratio; ALY, analysing methods; LSF, by least-squares best fit on lines; GCA, great circle analysis on planes; CMB, combined line and plane methods; C, coordinates: G, geographic; S, stratigraphic;

*, two specimens whose companion cores were used in calculating the LSF mean.

+, Recalculated on the basis of selected means given in the original publications.

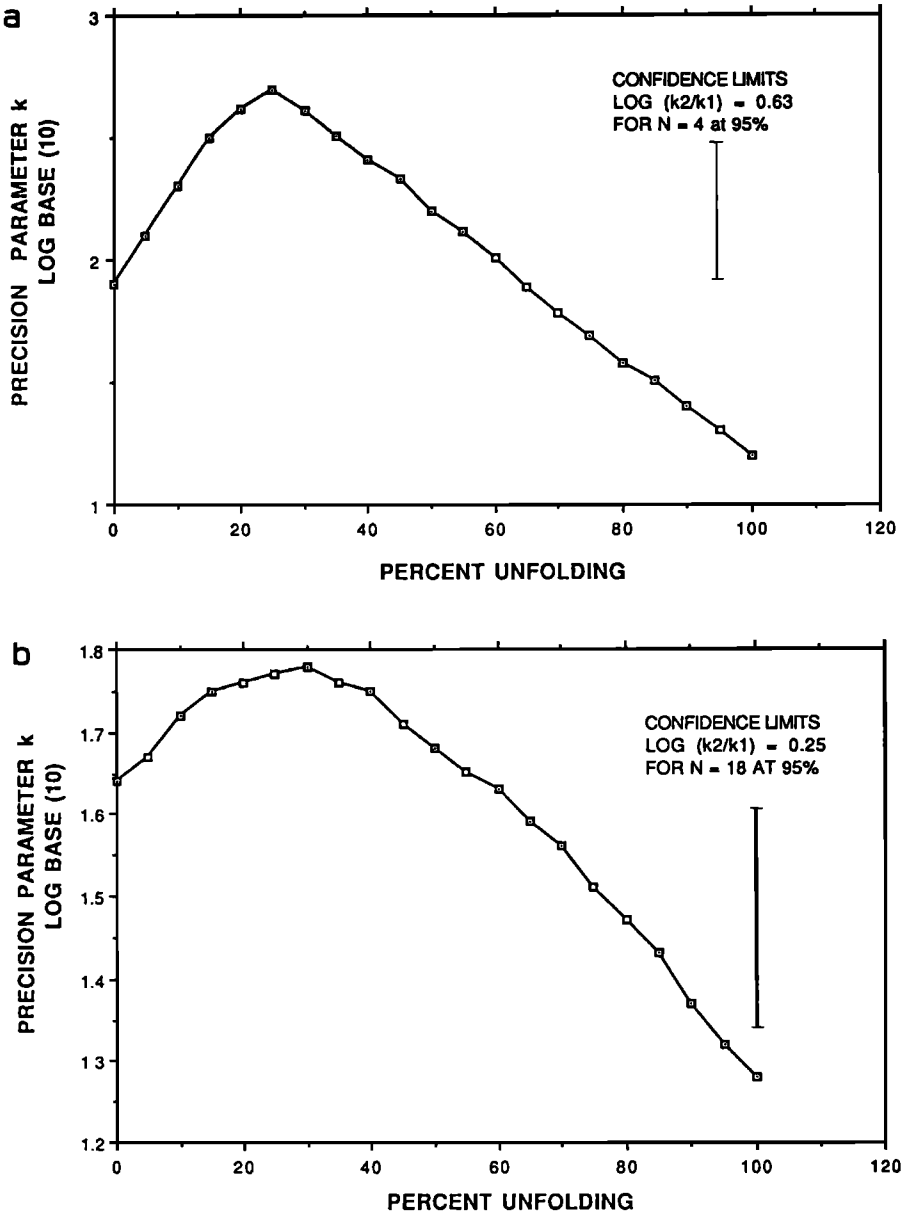


Fig. 7. Logarithm of Fisher precision parameter k values versus percent unfolding for the mean of the A component derived from (*a*) four sites from the Jingxin area (k values strongly peak at about 25% unfolding, and k at this unfolding step is significantly greater that most other tilt corrected k values at 95% confidence level) and (*b*) sites NCB.076 and NCB.077 (N = 18 samples) in Shandong province (the k values peak at about 30% unfolding, but are significantly greater than that at 0% unfolding at about 80% confidence only). The advantage of plotting k on a logarithmic scale is to allow the 95% confidence limit (vertical line) to be applicable over any portion of the curve.

samples finally struggling to reach the direction or by demagnetization trajectories of other samples that did not converge to the origin, but which trend toward the characteristic southerly quadrant on stereonet remagnetization planes. The ChRM component in these sites could not be adequately resolved by fitting least- squares lines, apparently because it is of smaller relative magnitude and has an unblocking temperature spectrum that overlaps more with other components. Therefore, the remagnetization plane method was applied. The remagnetization planes of these samples, however, were largely parallel so that no well defined intersection could be determined. Because no reliable stable end-point ChRM determinations ("set points", see *McFadden and McElhinny* [1988]) from each site can be pooled with the great circle data to overcome a potentially significant bias problem involved in great circle analysis (as noted previously by *Schmidt* [1985] and described more recently by *Thrupp et al.* [1991]), we do not use the results from the great-circle techniques for these samples.

Late Precambrian to Early Cambrian Site (NCB.067 of Table 1)

Indication of the ChRM component is also found from rocks in the latest Precambrian to Early Cambrian marl site NCB.067. There, both thermal and AF demagnetization of seven marl samples first removed a large A component with unblocking

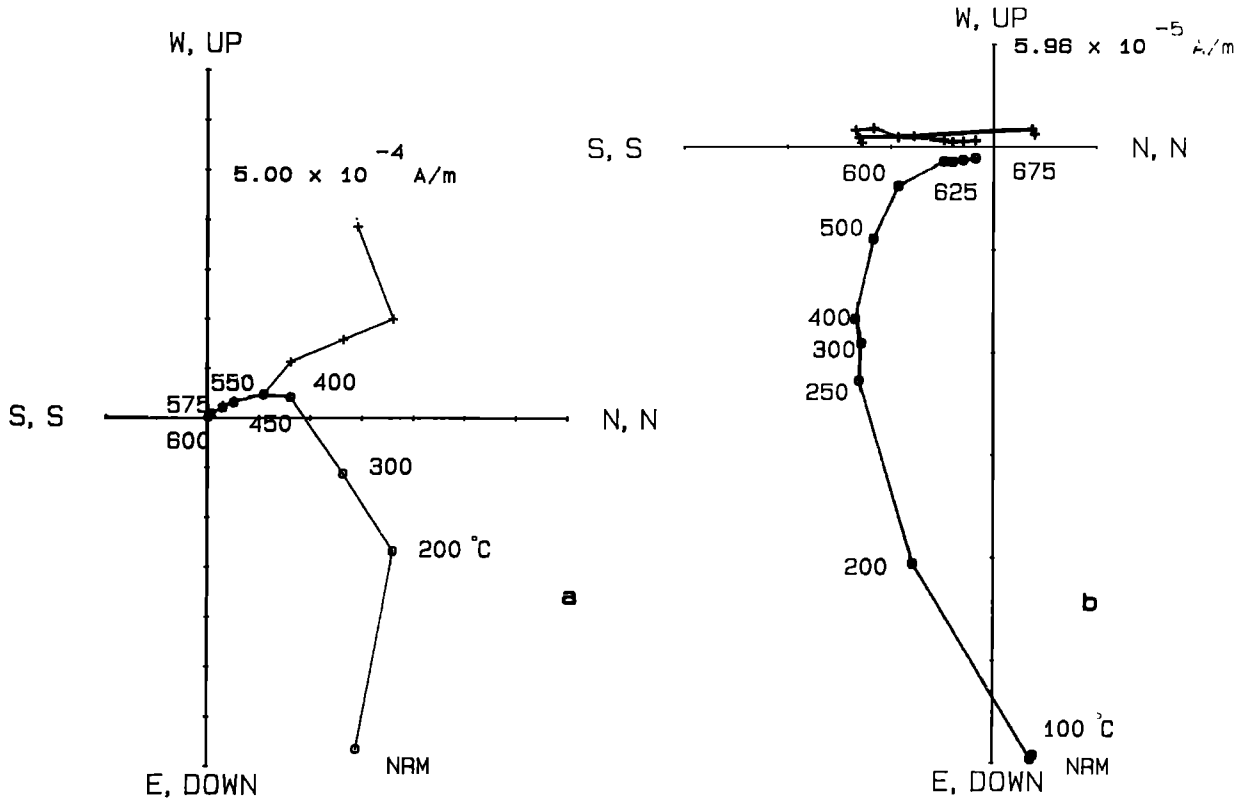


Fig. 8. Orthogonal vector plot of thermal demagnetization of Ordovician limestone samples in site NCB.080 in the Wennan area of Shandong province. (*a*) reversed ChRM component; (*b*) normal ChRM component. Directions are plotted in geographic coordinates. Crosses and open circles same as Figure 2.

temperatures of 500 to 650°C and coercivities of about 60 mT, and only after that thermal demagnetization revealed the ChRM component from six samples with unblocking temperatures of 675 to 700°C (two samples were analyzed by remagnetization plane techniques; see Table 4).

ANALYSIS AND DISCUSSION OF THE COMPONENTS OF MAGNETIZATION

Origin and Timing of the A *component*

In almost every site, the *A* component is well identified with low to intermediate unblocking temperatures. Combinations of demagnetization techniques suggest that the *A* component probably resides in magnetite in all cases (although components with unblocking temperatures less than 580°C do not preclude hematite). In some sites that are closely spaced, sometimes even within a site, the unblocking temperatures for the *A* component vary greatly. This would be very difficult to explain in terms of a simple thermal overprint other than a chemical remagnetization (or a combination of chemical and other processes), because, in the former case, the unblocking temperature should remain more or less the same. The absence of the *A* component in some sampling localities (e.g., the Nanliangdu locality in Jingxin area) is very hard to explain in terms of a regional overprint other than chemical, because a thermal overprint should affect all rock units similarly. The *A* component is characterized by northerly declination and moderate to steep inclination with both polarities, the majority of which are normal (Table 3). However, the normal and reversed polarities of the *A* component are never confined to stratigraphic horizons, but vary between closely spaced sites, sometimes even in the same core (see Figure 8). The most likely explanation for the observed random distribution of polarity is that the rocks carry a chemical remanent magnetization (CRM). Therefore, the detailed demagnetization behavior and the distribution of the normal and reversed directions clearly suggest that the *A* component reflects a chemical remagnetization.

The presence of reversed polarity of the *A* component in some rock units suggests that the remagnetization must have been imposed over a period of time that extends into pre-Brunhes time (>730,000 years). In the Xishan area, the in-situ site mean directions of the *A* component are statistically indistinguishable from that of the Cretaceous lamprophyre dike and the baked limestone (Table 2), indicating that they may be of the same age. The synfolding results at the Jingxin and central Shandong areas as well as other statistically significant synfolding signatures elsewhere in north and south China (as documented in detail in *Zhao* [1987]), add evidence to indicate that the *A* component was probably imposed during the major episode of regional deformation in the Late Mesozoic.

In light of the uncertainty in the age of the *A* component magnetization, it would be useful to compare the *A* component with other Late Mesozoic paleopoles of Chinese blocks. As shown in Table 3 and Figure 9, the paleomagnetic pole positions of the *A* component from the three areas of this study are

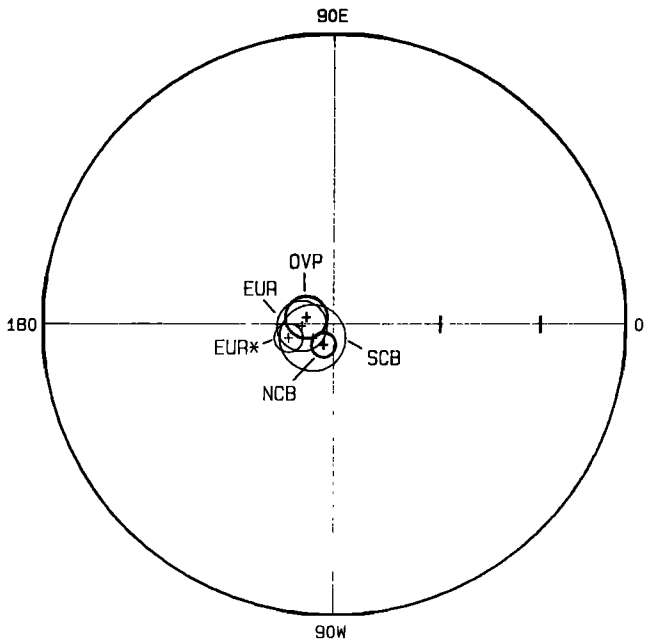


Fig. 9. Equal-area stereographic projection of Late Cretaceous paleomagnetic poles for Chinese and Eurasian blocks (NCB, SCB, and EUR) and the averaged virtual geomagnetic poles derived from the *A* component (OVP) from the three sampling areas (see Table 3). Triangle represents the mean paleopole of 70 - 100 Ma of *Besse and Courtillot* [1991]. Circles are the associated 95% confidence limits.

statistically indistinguishable from those of Late Cretaceous age for the NCB and SCB [*Kent et al.*, 1986; *Enkin et al.*, 1991; *Zhao et al.*, 1990; *Chan*, 1991] and of Late Mesozoic age for the Eurasian plate [*Lee et al.*, 1987; *Westphal et al.*, 1986; *Van der Voo and French*, 1974; *Besse and Courtillot*, 1991]. The data (Table 3 and Figure 9) therefore suggest that these three distinct sampling areas, which are presently some 300 km apart, record a Late Mesozoic remagnetization of regional scale. The age of this remagnetization appears to be coincident with the Late Mesozoic thrusting/deformation in the interior of east China [*Hsü et al.*, 1990], which in turn was closely related in space and time with the period of anomalously high convergence and subduction between the Kula and Pacific plates in the Late Cretaceous [*Weng and Wang*, 1981; *Huang*, 1984; *Engebretson et al.*, 1984, 1986; *Rea and Duncan*, 1986]. This CRM overprint may have been caused by the migration of tectonic fluids [*Oliver* 1986; *McCabe et al.*, 1987] that were expelled from the Yanshanian orogenic belt as a result of these tectonic events.

Origin and Timing of the ChRM Component

The thermally discrete high unblocking temperature component was successfully isolated in samples from 18 sites (Table 4) distributed over the three widely separated sampling areas. These rock samples are of various lithology (see Table 1) and were measured on different magnetometers in different laboratory environments, years apart. Yet all revealed the same ChRM component, which suggests that this component is a true paleomagnetic signature in these rocks.

To distinguish the ChRM components from various rock units, we applied the F-test of *McFadden and Jones* [1981] to the mean direction of each site from the three sampled areas. The F-test results suggest that the mean directions for Early to Middle Cambrian and Late Cambrian to Ordovician sites are distinct at 95% confidence level in each area. On the other hand, we noticed in the literature [*Armstrong*, 1978; *Cowie and Johnson*, 1985; *Harland et al.*, 1982; *Palmer*, 1983; *Snelling*, 1987; *Odin*, 1982; *Odin et al.*, 1983; *McKerrow et al.*, 1985] that the possible age range for the Precambrian-Cambrian boundary is 530-600 million years (Ma) before present; for the Middle-Late Cambrian boundary, 523-530 Ma; for the Cambrian-Ordovician boundary, 495-523 Ma; and for the Llanvirn-Llandeilo (Middle Ordovician) boundary, 450-470 Ma. Thus, according to the F-test results and the recent reassessments of the radiometeric dates mentioned above, it would seem more logical to group the Early-Middle Cambrian and Late Cambrian to Middle Ordovician data from each sampling area separately (Cambrian and Ordovician in Table 4). It may be worth mentioning that the choice of grouping Late Cambrian and Early Ordovician data together has frequently been made in the literature with little explanation of underlying rationale [e.g., *McElhinny et al.*, 1977, 1979; *Piper*, 1988]. In Shandong, the Late Precambrian-Early Cambrian site spanned the boundary between the Late Precambrian and Early Cambrian, and the polarity record of the samples compares well with that of the formations that are equivalent in age but from widely separated localities. For these reasons we include the ChRM component of this site in the final mean calculation for the Cambrian paleomagnetic pole.

Tilt and fold tests. As listed in Table 4, the local tilt test on both Cambrian and Ordovician means yields inconclusive results for most sites. However, the Cambrian means for the Jingxin area and the Hebei sampling areas as a whole (Jingxin and Xishan) pass the tilt/fold tests. The F-ratio criteria indicate that the tilt/fold tests are positive at more than 90% and 99% confidence, respectively, indicating that the ChRM component isolated from Hebei is prefolding.

The regional fold tests are all positive, demonstrating the prefolding age for the ChRM component. As indicated in Table 4 and Figure 10, the best estimate (K) of Fisher's precision parameter for the mean VGP position of the 10 Cambrian sites before and after tectonic correction is 3.5 and 16.2, respectively. For the mean VGP of the eight Ordovician sites, K is 3.8 before correction, and 21.1 after tectonic correction. Applying the classical fold test [*McElhinny*, 1964], the increase in K corresponds to a positive fold test at more than 99% confidence level for both the Cambrian and Ordovician means. The fold test of *McFadden and Jones* [1981], in light of the more conservative nature of McElhinny's test, would simply reinforce the level of significance of these positive fold tests.

Application of an incremental fold test confirms that the fold tests are entirely positive; there is no occurrence of K-peak during incremental unfolding. Hence, both Cambrian and Ordovician ChRM components are prefolding. The age of the oldest folding that affected the sampled strata for all three areas, however, is no older than Late Triassic (Indosinian orogeny) and may be as young as Late Cretaceous. Thus the positive fold test results do not furnish a strong constraint on the age of magnetization. Other evidence is needed to help bracket the age of the components of remanent magnetization.

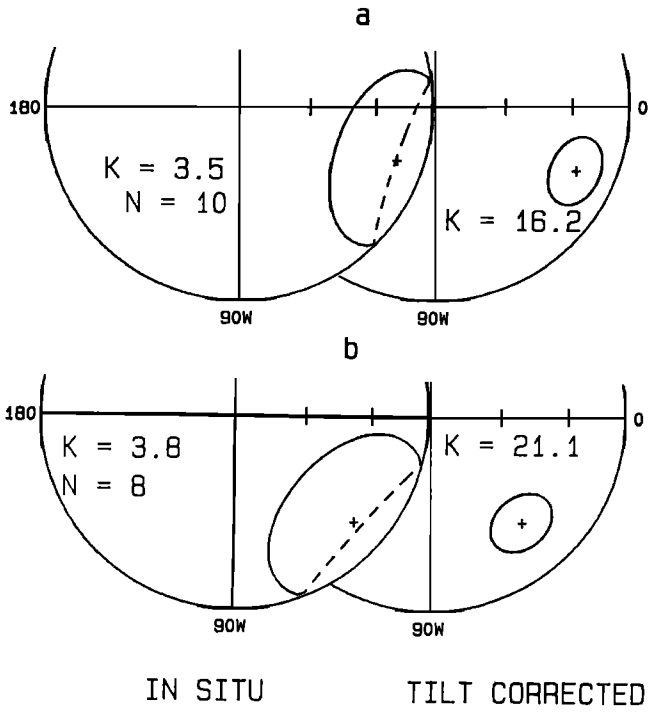


Fig. 10. Equal-area plot of mean VGP of the ChRM component of the (*a*) Cambrian and (*b*) Ordovician rocks from the NCB examined in this study. The data pass the regional fold test at more than 99% confidence, as indicated by a factor of 4.6 and 5.6 times increase in Fisher's precision parameter K, respectively (see Table 4). Crosses and solid boundaries on confidence cones indicate lower hemisphere projection; dashed boundaries denote upper hemisphere projection.

Reversal test. The dual-polarity distribution is apparent for the in-situ site mean directions and is very close to antiparallel, suggesting the ChRM directions are not seriously contaminated by some noncoaxial magnetization (see Figure 11 for an example). The F-test on the mean directions of normal and reversed polarities for both Cambrian and Ordovician sites yields positive results: the two polarity directions are not statistically different from each other at the 95% confidence level if the mean direction of one polarity is inverted to the other polarity. The successful reversal test proves that most secondary components have been removed and adds credence to the hypothesis that the ChRM may be primary.

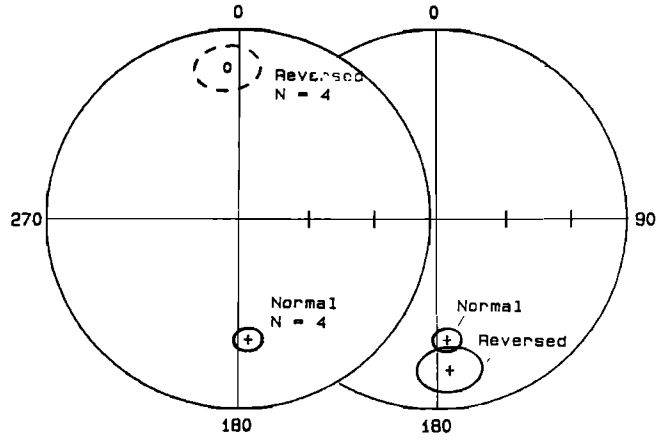


Fig. 11. Mean directions of the ChRM component of two Ordovician sites from Shandong province (Table 4), showing positive reversal test results. Directions are plotted in stratigraphic coordinates. Open circles and dashed boundaries on confidence cones denote upper hemisphere projection; crosses and solid boundaries indicate lower hemisphere projection.

Magnetic polarity pattern. Although the presence of reversals suggests that the acquisition of the ChRM components for both Cambrian and Ordovician rock units spanned considerable time, it is by no means a proof of primary magnetization. As discussed already, the Late Mesozoic chemical remagnetization (the *A* component) also exhibits dual polarity that is antipodal. Although the positive results of the baked contact, fold, and reversal tests support an early origin for this component, the age of magnetization is still not well constrained.

The magnetic polarity patterns between these three localities, on the other hand, seem to provide an alternative method of determining whether the strata were magnetized at or near the time of their deposition, hence helping to decide whether the ChRM is primary or near-primary remanent magnetization or not. Despite the wide geographical separation, the fossil assemblages and other geological indicators permit the correlation of formations sampled in the three localities. If the dual polarity magnetization is penecontemporaneous with deposition, the normal and reversal polarity patterns within a stratigraphic zone should agree reasonably well from locality to locality. On the other hand, if the component was overprinted in an irregular fashion at some later time during diagenesis, it would be unlikely for similar patterns to emerge. Indeed, we have not observed any polarity correlations between parallel sections for overprinted components which are of dual polarity in this study or in our previous studies.

As shown in Figure 12, there is an overall similarity in the magnetic polarity pattern of the ChRM component in the stratigraphic sequence at the three localities. This general agreement lends support to the hypothesis that the ChRM component was acquired at or shortly after deposition of the formations in these localities. At this point, we emphasize that the term of magnetic polarity patterns used here is an example of "elastic application." The polarity column in Figure 12 is by no means as detailed a pattern as is used in magnetostratigraphic studies. Even so, an attempt was made to compare this polarity column with records published for other parts of the world, such as the Meishucun section in South China [*Fang et al.*, 1988-1989], Salany-Gol section in Mongolia [*Kirschvink et al.*, 1987], Lena River section in Siberia [*Kirschvink and Rozanov*, 1984], Amadeus Basin in Australia [*Kirschvink*, 1978], Johnnie Formation in the Desert Range of Nevada [*Gillett and Van Alstine*, 1979], and Lie de Vin Formation of Morocco [*Ripperdan*, 1990]. All records include frequent reversals. However, due to age differences and the possible presence of disconformities, hiatuses, and remagnetization in some of the sections, it is difficult to make any detailed correlations between our polarity column and those sections at this time.

SUMMARY AND COMPARISON

In summary, our data revealed multicomponents of magnetization from three widely separated localities in the NCB.

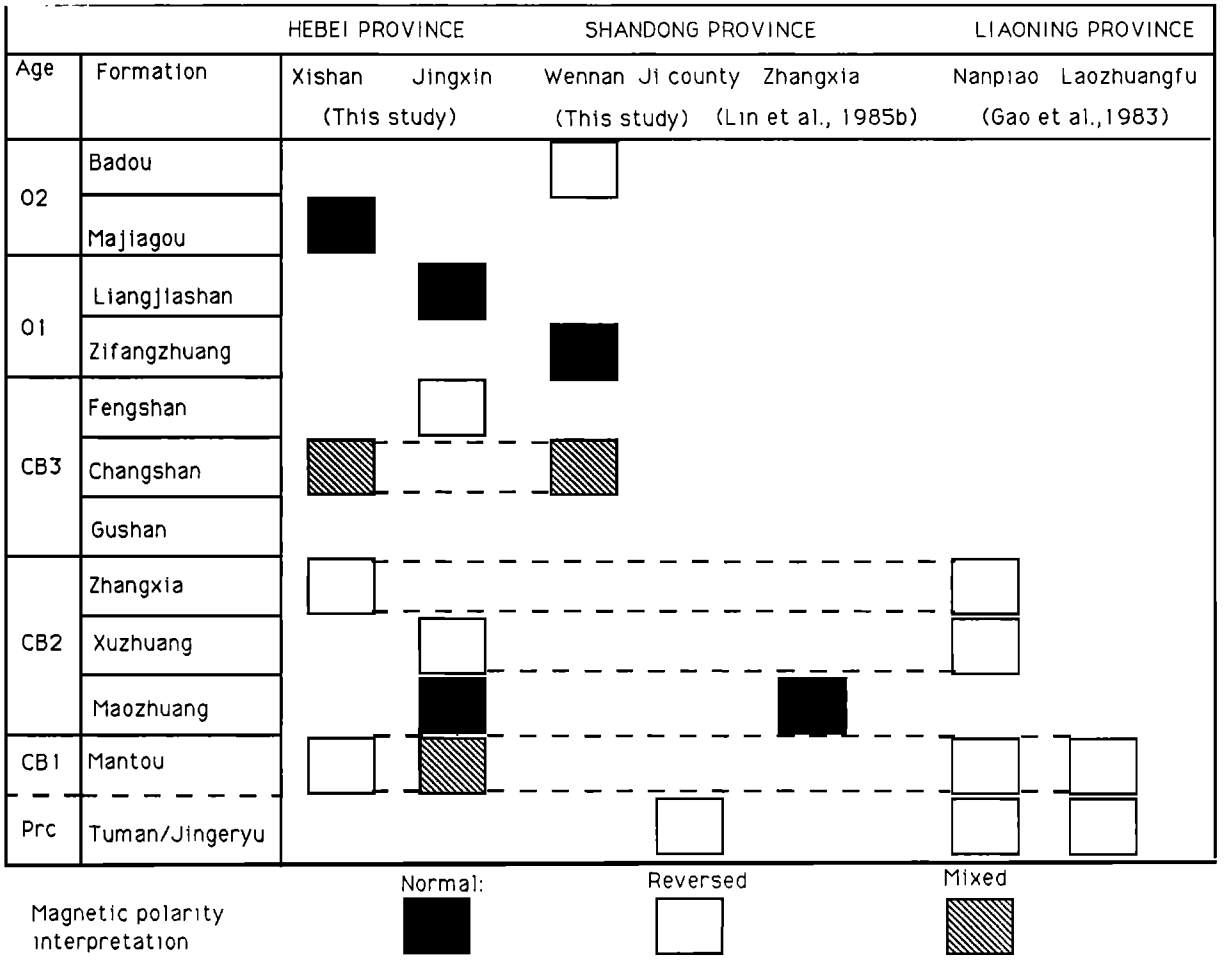


Fig. 12. Sketch plot to show the comparison of magnetic polarity zonation at four areas sampled in this study and three sections by *Lin* [1985*b*] and *Gao et al.* [1983]. Magnetic polarity interpretation: black, reversed polarity; white, normal polarity; and stripes, mixed polarity.

The high temperature components were isolated after removal of a large overprint. Because of the linear final demagnetization curves, the internal consistency of mean directions from rocks of different lithology from widely distributed localities, the positive results of site mean reversal, baked contact, and fold tests, the overall similarity of normal and reversed distribution patterns in the stratigraphic section, plus the fact that the magnetization is carried by both magnetite and hematite, which are capable of preserving the early remanent magnetization over a long period of time, we conclude that the high temperature components are the characteristic components and represent records of the paleomagnetic field close in age to the deposition of these Cambrian and Ordovician rocks. In fact, our data satisfy all seven reliability criteria recently proposed by *Van der Voo* [1989, 1990]. Thus they are suitable for calculating Cambrian and Ordovician poles for the NCB.

We have done this in two ways: (1) averaging by regions and (2) averaging by sites. Both methods yield essentially the same results, but we prefer the method of averaging by sites because the mean directions are more comparable in terms of numbers of samples. The overall average VGP of the 10 Cambrian sites corresponds to a paleopole at longitude 335.2°E, latitude 21.2°N (A_{95} = 12.4°, K = 16.2). For the eight Ordovician sites, the pole position is longitude 310.9°E, latitude 28.8°N, A_{95} = 12.3°, K = 21.1 (Table 4).

In order to determine whether these paleomagnetic pole positions are north or south poles, one has to know the geomagnetic polarity for the times when the components of magnetization were recorded. Because of the scarcity of reliable paleomagnetic data in China for most of the Paleozoic (except Late Permian time), it is impossible to extrapolate back in time from paleomagnetic data to constrain the polarity for the Early Paleozoic.

The predominantly negative inclination of our Cambrian site direction can be interpreted to indicate that the NCB lay either in the southern hemisphere during a mainly normal polarity period or in the northern hemisphere during a mainly reversed polarity period. The two options for the Cambrian position of the NCB north paleopole are (1) in the Atlantic or (2) in the south Pacific near the east coast of Australia. Option 1 places the sampling area of the NCB in the southern hemisphere at about 23°S, whereas option 2 implies that it was at 23°N in the northern hemisphere. Although it can be argued that both options yield the same conclusion (the NCB was within equatorial areas), we favor the

second option, mainly for reasons involving magnetic polarity bias.

Almost all available data (APW path reconstructions, paleomagnetic studies on reference cores, and magnetostratigraphic results) of Cambrian and Ordovician rocks in North America, Europe, Africa, and Australia suggest that the Earth's field was mainly of reversed polarity throughout Early Paleozoic time [*Elston and Bressler*, 1977; *Gillett and Van Alstine*, 1979; *Kirschvink and Rozanov*, 1984; *Khramov and Rodionov*, 1981; *Khramov et al.*, 1985; *Kirschvink and Ripperdan*, 1986; *Krs et al.*, 1987; *Perrin et al.*, 1988; *McElhinny and Embleton*, 1974; *Klootwijk*, 1980; *Li et al.*, 1990; *Bachtadse and Briden*, 1990; *Kent and Van der Voo*, 1990; *Kirschvink et al.*, 1991]. If this generalization is true, then the dominantly negative inclination of the ChRM components found in this study would place the NCB in the northern hemisphere. This polarity designation is used in Table 4. It may turn out eventually that this polarity decision needs to be switched, but such a reinterpretation would not diminish the tectonic value of the data. For the time being, we will use these poles to compare the existing poles that are relevant to this study.

There are three previously published Cambrian paleomagnetic poles available for the NCB (see Table 4). One of them is a study of three Early to Late Cambrian red oolitic dolomite and limestone formations from Shandong (site longitude 116.9°E, latitude 36.5°N) and Liaoning (site longitude 121.7°E, latitude 39.4°N) provinces [*Lin et al.*, 1985*a*, *b*; *Lin and Fuller*, 1990]. The reported pole position (longitude 298.6°E, latitude 15.0°N, A_{95} = 9.9°) lies in the same quadrant as the pole for this study, but the F-test results show that they are different at greater than 99% confidence (Table 4 and Figure 13). One of the possible explanations for the disagreement could be the age differences between the two Cambrian poles. As described, the Cambrian pole obtained from this study is derived from Early to Middle Cambrian rock formations, whereas *Lin and Fuller* [1990] comprised six sites (35 samples) ranging from Early to Late Cambrian as well as the mean of one site (23 samples) of Early Cambrian sandstone from the study of *Gurarii et al.* [1966] in North Korea [see *Lin*, 1984, p. 18], for which there were no fold or reversal tests. Indeed, we noticed that the Cambrian pole of *Lin and Fuller* [1990] falls closer to our Ordovician pole (Figures 13 and 14).

Nevertheless, despite the differences in pole locations, it is very interesting to note that the polarity records from one of our Cambrian sites in the Jingxin area in Hebei agree very well with that from the same formation (Maozhuang formation) studied by *Lin et al.* [1985*b*] in Shandong (see Figure 12). This reinforces the interpretation that the ChRM component found in our study is possibly primary.

Another published paleomagnetic study [*Gao et al.*, 1983] is on Late Precambrian to Ordovician carbonates from east and west Liaoning province in the NCB. The reported results are provisional; the number of samples are generally small and some samples were only treated with blanket demagnetization techniques. Reanalysis of the data given in the original publication reveals that rocks from the east Liaoning sites probably suffer from the problem of remagnetization and have been

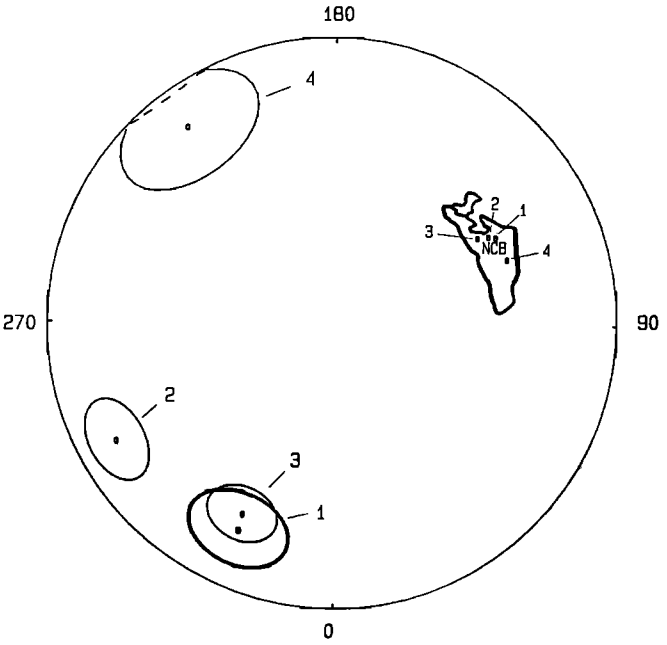


Fig. 13. Equal-area plot of NCB Cambrian paleomagnetic poles with A_{95} confidence limits and approximate positions of the sampling areas in the NCB. 1, longitude 335.2°E, latitude 21.2°N, A_{95} = 12.4°, this study; 2, 298.6°E, 15.0°N, A_{95} = 9.9° [*Lin and Fuller*, 1990]; 3, 334.5°E, 26.8°N, A_{95} = 8.9° [*Gao et al.* 1983]; 4, 217.2°E, 15.3°N, A_{95} = 17.5° [*Wu*, 1988]. For the purpose of clarity, all the poles are plotted with reversed polarity. Longitudes and latitudes for the corresponding sampling sites are 1 [118.0/36.0]; 2 [119.3/37.9]; 3 [120.7/41.1]; and 4 [110.2/35.5]. Crosses and solid boundaries on confidence cones indicate lower hemisphere projection; dashed boundaries denote upper hemisphere projection.

incorrectly interpreted as a primary component of magnetization. We have recalculated a pole from the best data for five of their sites (40 samples) from two localities in east Liaoning. Choosing consistent polarity and applying the F-test of *McFadden and Jones* [1981], this recalculated pole (longitude 334.5°E, latitude 26.8°N, A_{95} = 8.9°) agrees very well (not significantly different at 95% confience level) with our present study (Table 4 and Figure 13).

It is interesting to note that comparison of the mean declinations of sampling sites in Liaoning and Hebei provinces that are presently located on the east and west sides of the Tancheng-Lujiang fault (Figure 1) suggests no paleomagnetically detectable rotations for this fault since the remanence was acquired. The inclinations for the two sections in Liaoning are all negative, which is in excellent agreement with those for the Early Cambrian samples in the Xishan and Jingxin areas (Figure 12). The "global" similarity in the polarity patterns once again suggests that the ChRM component in these Early Cambrian strata is a primary remanence or a very early CRM. As for the Precambrian formations, we would like to point out that, to our knowledge, there is no strong evidence to show that the Jingeryu formation at Liaoning province may be much older that the Tumen formation at Shandong. If it is, the polarity comparison for these Precambrian formations (Figure 12) is fortuitous.

The third reported Cambrian pole came from the Hanchen area (longitude 110.2°E, latitude 35.5°N) of Shaanxi province [*Wu*, 1988], which was derived from 20 Early Cambrian red marl

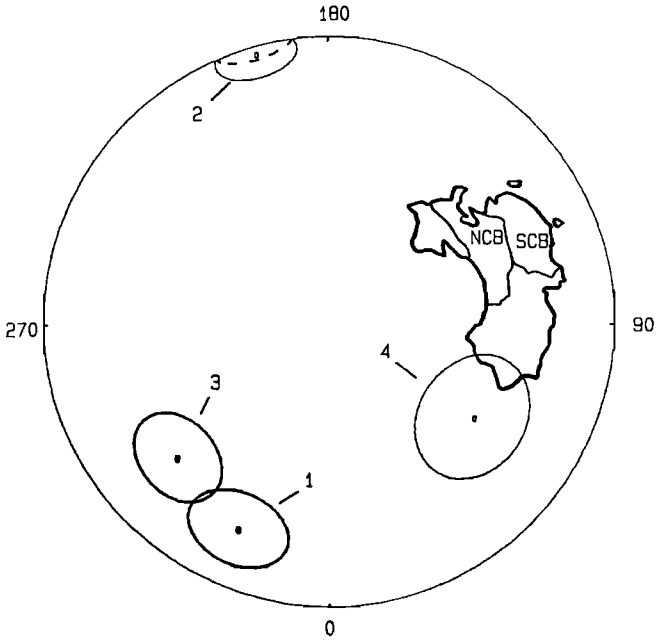


Fig. 14. Simplified tectonic map and Cambrian and Ordovician paleomagnetic poles and 95% confidence circles for NCB and SCB. 1, Cambrian pole for NCB, longitude 335.2°E, latitude 21.2°N, A_{95} = 12.4°, this study; 2, Cambrian pole for SCB, 195.0°E, 3.4°N, A_{95} = 8.8° [*Lin and Fuller*, 1990]; 3, Ordovician pole for NCB, 310.9°E, 28.8°N, A_{95} = 12.3°, this study; and 4, Ordovician pole for SCB, 55.7°E, 38.9°N, A_{95} = 16.9°. For the purpose of clarity, all the poles are plotted with reversed polarity. Crosses and solid boundaries on confidence cones indicate lower hemisphere projection; dashed boundaries denote upper hemisphere projection.

samples and falls at 217.2°E, 15.3°N, with A_{95} = 17.5°. Our Cambrian pole is very different from this Early Cambrian pole (Table 4 and Figure 13). Unfortunately, the results of *Wu* [1988] for the Early Cambrian are the average of two markedly different distributions of component directions (southwesterly down and southeasterly up). The southeasterly up direction group corresponds to a pole falling very close to the Late Permian pole for the NCB [*Coe et al.*, 1987; *Zhao and Coe*, 1989; *Meng et al.*, 1989], suggesting that a Late Permian remagnetization of some of the samples cannot be precluded.

Lin et al. [1985*a*] also reported a Mid-Ordovician pole from five rock samples at one limestone site in Liaoning province (site longitude 121.7°E, latitude 39.4°N). The result (longitude 332.5°E, latitude 43.2°N, A_{95} = ?) is again statistically different from the Ordovician poles obtained in this study (at more than 95% confidence). Because no details concerning demagnetization behavior are given for this work, which comprised a very small number of samples from only one site, this result cannot be considered a representative pole and has recently been passed over by the original authors [*Lin and Fuller*, 1990].

Comparison With Cambrian Poles From South China Block (SCB)

Four Late Precambrian-Early Cambrian paleomagnetic poles are available for the SCB (Table 5). Three of them are from a Late Precambrian-Early Cambrian sequence of carbonates and phosphorites (the Meishucun Formation) from two localities in Yunnan province [*Lin et al.*, 1985*b*; *Liu et al.*, 1988; *Fang et al.*, 1988-1989]. The results are completely different from each other, and also different from our Cambrian pole for the NCB. None of the three studies, however, can demonstrate the age of magnetization with certainty.

The fourth Early Cambrian pole [*Lin et al.*, 1985*b*; *Lin and Fuller*, 1990] was derived from rocks in Zhejiang, Hubei, and Yunnan provinces that have both polarities and pass a generalized tilt test (Table 5). This pole is very different from the Cambrian pole of the NCB, implying that North and South China did not have their same relative orientations in the Early Paleozoic (Figure 14). It may be worth commenting that the Cambrian pole position for the SCB falls close to the Silurian paleomagnetic pole (see Table 5). The similarity of these two pole positions could imply either that there was not much polar wander of the SCB during this extended period of time or, as suggested by some workers [*Opdyke et al.*, 1987; *Fang et al.*, 1989], that the Cambrian rocks were remagnetized in the Middle Paleozoic.

A provisional Mid-Late Ordovician pole (235.7°E, 38.9°S, N = 5 sites, A_{95} = 16.9°) for the SCB has recently been reported in a short article [*Fang et al.*, 1990*b*], which was derived from limestone, mudstone, shale, and sandstone from one area in Yunnan province. Although the authors acknowledge that the magnetization might be an overprinted directions, they interpret the pole as representative of the Ordovician geomagnetic field in South China. This pole is completely different from that of the NCB (Figure 14), and implies a relatively high paleolatitude (48.5°) for SCB in Ordovician time. The hemisphere is indeterminate because of the ambiguity in polarity, but *Fang et al.*'s [1990*b*] preference places the SCB in the southern hemisphere near East Africa in Ordovician time.

As far as we can ascertain, there is still no reliable Cambrian and Ordovician paleomagnetic data for other blocks and terranes in China, underscoring the need for further paleomagnetic studies of Early Paleozoic rocks from these regions.

TECTONIC AND PALEOGEOGRAPHIC IMPLICATIONS

Although the results presented in this paper are limited, it is useful to speculate on the tectonic and paleogeographic implications of the data. The paleomagnetic data obtained in this study suggest that the NCB occupied low latitudes during both Cambrian and Ordovician times, quite far away from today's position. These data are in good agreement with the paleobiogeographical reconstructions of China for Early Paleozoic time [*Wang*, 1984; *Cocks and Fortey*, 1990; *Bergström*, 1990; *Crick*, 1990; *Berry and Wilde*, 1990] as well as the paleozoogeographic patterns of brachiopods for Late Ordovician time [*Sheehan and Coorough*, 1990]. In addition, the extensive and thick Early Paleozoic carbonates and phosphate deposits of the NCB also support the paleomagnetic evidence. These paleoclimatic indicators suggest that the NCB was situated in a low-latitude area of tropical climate [*Cui and Zhen*, 1984]. Thus, the close agreement between paleomagnetic and paleobiologic data of the NCB offers a coherent picture of Early Paleozoic paleogeography for the block.

TABLE 5. Published Paleozoic Paleomagnetic Poles and Paleolatitudes for Major Chinese Blocks and Terranes

Age	SLong/SLat	PLong/PLat	Paleolatitude	Reference
		North China Block (Including Inner Mongolia)		
P2	112.4/37.8	358.5/44.0	11.3	*McElhinny et al.* [1981
P2	112.4/37.8	358.5/49.2	14.8	*Zhang* [1984]
P2	112.3/37.8	6.3/38.1	11.9	*Lin* [1984]
P2	113.8/37.9	6.2/56.7*	22.5	*Lin et al.* [1984]
P2	97.8/39.8	351.3/43.9	16.7	*Coe et al.* [1987]
P2	114.4/37.5	1.4/55.2	18.8	*Zhao and Coe* [1987]
P2	112.1/38.6	2.9/37.9	10.4	*Zhao and Coe* [1987]
P2	113.2/40.1	335.7/55.2	12.0	*Zhao and Coe* [1987]
P2	109.5/34.5	13.7/42.3	18.6	*Cheng et al.* [1988]
P2	118.9/43.0	339.5/52.5	11.7	*Zhao et al.* [1990]
P2	110.2/42.3	355.6/56.7	23.2	*H. Liu* [1987]
C3	120.1/43.9	338.1/32.6	-6.0	*Zhao et al.* [1990]
C3	110.2/35.5	11.9/30.0	10.9	*Wu* [1988]
O	118.0/36.0	310.4/28.8	24.1	*This study*
CB2	110.2/35.5	217.2/15.3	-4.4	*Wu* [1988]
CB	120.7/41.1	334.5/26.8*	28.4	*Gao et al.* [1983]
CB	118.0/36.0	334.7/21.3	23.0	*This study*
CB1	119.3/37.9	298.6/15.0	34.9	*Lin and Fuller* [1990]
		Hexi Corridor Terrane		
CB2	105.0/37.0	228.6/48.0	8.7	*He et al.* [1988]
		North Qilian Mts. Terrane		
O1	104.0/36.5	353.0/32.7*	4.6	*Y. Li et al.* [1987]
		Chaidam Terrane		
CB2	96.0/37.0	334.0/52.3	2.6	*Y. Li et al.* [1986]
		Tarim Block		
P2	83.4/42.1	188.0/72.0	35.4	*McFadden et al.* [1988]
P2	78.8/39.8	181.0/66.0	31.2	*Y. Li et al.* [1988]
P1	79.3/40.7	172.0/55.0	30.9	*Sharps et al.* [1989]
C3	78.8/40.5	160.0/41.0	30.9	*Zhai et al.* [1987]
D3	78.8/40.5	151.2/-10.5	6.2	*Bai et al.* [1987]
D	78.8/40.5	165.0/16.0	13.2	*Y. Li* [1990]
		Kunlun Terrane		
C	95.0/36.0	223.9/13.0	-21.3	*Lin and Watts* [1988]
		South China Block		
P2-T1	105.5/32.4	225.5/47.7	6.4	*Heller et al.* [1988]
P2-T1	106.4/32.4	223.6/46.1	6.8	*Steiner et al.* [1989]
P2	103.4/29.6	252.0/54.0	-2.1	*McElhinny et al.* [1981]
P2	103.4/29.6	252.0/49.7	-5.9	*Zhao and Coe* [1987]
P2	103.4/29.6	241.8/54.1	1.2	*Chan et al.* [1984]
P2	103.4/29.6	248.3/50.9	-3.8	*Zhang* [1984]
P2	103.0/25.0	241.0/50.0	-6.3	*Fang et al.* [1990*a*]
P2	103.5/29.7	244.2/54.4	-0.7	*C. Liu et al.* [1987]
P2	103.0/28.3	251.3/54.3	-3.0	*Zhou et al.* [1986]
P2	102.9/26.7	226.2/49.5	1.3	*Huang et al.* [1986]
P2	105.7/26.4	235.3/29.3	-16.3	*J. Lin* [1984]
C	120.2/30.2	223.8/22.2	-0.1	*J. Lin et al.* [1985*a*]
D2	103.0/24.5	190.4/-8.9	-1.3	*Fang et al.* [1989]
D1	119.0/32.0	277.5/57.2	1.0	*Tan and Zhai* [1988]
S	108.0/28.5	194.7/4.9	5.2	*Opdyke et al.* [1987]
O	103.0/24.5	55.7/38.9	48.5	*Fang et al.* [1990*b*]
CB	102.3/24.4	95.0/3.4	-1.1	*Lin and Fuller* [1990]
Prc	102.7/24.5	270.7/68.8	3.7	*Fang et al.* [1988-1989]
Prc	102.3/24.4	211.1/7.1	13.9	*J. Lin et al.* [1985*b*]
Prc	102.7/24.5	226.7/74.2	15.1	*Liu et al.* [1988]

Age, Prc, Precambrian, CB, Cambrian, O, Ordovician, S, Silurian, D, Devonian, C, Carboniferous, P, Permian, T, Triassic; 1, Early, 2, Middle, 3, Late. SLong/SLat(PLong/PLat), East longitude and north latitude of sampling area (pole position).

*, Recalculated on the basis of selected means given in the original publications.

With regard to global paleogeographic reconstructions, it is still a very difficult task to reconstruct the Paleozoic because the amount of reliable paleomagntic data, which are the only source of quantitative data on the ancient orientations of the continents, are sparse. Many suggestions have been tendered concerning the locations of the various Chinese blocks and terranes during the Paleozoic: part of Gondwana [*Kirschvink*, 1991], part of Laurasia [*Zh. Zhang et al.*, 1984], part of more hypothetical ancient

continents such as Pacifica [*Nur and Ben-Avraham*, 1977] or Cimmeria [*Sengör*, 1984], or simply standing alone [*Ziegler et al., 1979*; *Fang et al.*, 1989]. Paleozoic paleomagnetic data for China are, at present, insufficient to resolve such questions concerning Paleozoic paleogeography. However, almost all of the published paleomagnetic results (not necessary all reliable) show that the major blocks of China were at equatorial latitudes both early and late in the Paleozoic, as summarized in Table 5. The consistently low paleolatitude (all but one within 35° of the equator) of the various Chinese blocks is striking and may have tectonic or paleomagnetic implications. Whether it implies that these blocks were confined to low latitudes throughout the entire Paleozoic or that it is a result of widespread Late Paleozoic remagnetization is not clear at present.

In a very recent paleogeographic model, *Kirschvink* [1991] has placed both North and South China attached to Australia for the entire Cambrian period. This reconstruction is at odds with the very significant differences in the APW path for Australia [*Li et al.*, 1990] and that of this study for North China during the Cambrian. Moreover, he notes that data from several continents suggest apparent polar wander of around 90° during this time, and concludes that 45° of this might represent true polar wander. Our data, however, indicate only about 30° of apparent polar wander for the NCB from Early Cambrian to Middle Ordovician (Table 4 and Figure 14), a considerably longer interval. Clearly, more data are needed from Early Paleozoic formations of North China to test the robustness of this discrepancy.

In any case, we know by Late Paleozoic time both North and South China were not part of Gondwana, because they were in the equatorial latitudes and in the initial stages of collision and amalgamation with proto-Asia [*McElhinny et al.*, 1981; *Opdyke et al.*, 1986; *Zhao et al.*, 1989, 1990]. On the other hand, most parts of Gondwana were still at polar latitudes during the Late Paleozoic [*Powell and Veevers*, 1987; *Scotese and Barrett*, 1990; *Li et al.*, 1990]. Although recent geologic investigation suggests that parts of the SCB were separated from Gondwanaland during the Late Precambrian [*Hsü et al.*, 1990], paleontological data from South China show affinity with Australia in both the Cambrian [*Burrett and Richardson*, 1980] and Devonian [*Long and Burrett*, 1989], which is consistent with the paleomagnetically inferred low latitudes and suggests that at least South China may well have been part of Gondwana in Early and Middle Paleozoic time. Whether or not North China belonged to Gondwana during this same period is less clear, but *Burrett et al.* [1990] recently suggested that Cambrian and Early Ordovician fauna from North China are similar to those of Australia.

If, as seems likely from paleomagnetic and geological evidence from India, Tibet, and southeast Asia, and from analysis of marine geophysical data from the Indian Ocean, Asia grew by accretion of fragments rifted from Gondwana, the NCB and SCB may have been the first such pieces to go. Our work on testing this hypothesis on newly discovered Silurian and Devonian rocks from Ningxia, Early Permian rocks from Shaanxi of the NCB, and Devonian and Carboniferous rocks from Guangxi of the SCB is currently in progress, and may provide information on the timing of rifting. This is needed to evaluate the geodynamic processes envisioned by *Anderson* [1982], in which the accumulation of heat under the supercontinent is thought to have caused it to shift from the pole and break into pieces.

CONCLUSION

Two components of magnetization can be isolated by extensive thermal demagnetization on several Cambrian and Ordovician sedimentary units from localities in Hebei and Shandong provinces of the North China Block. The lower unblocking temperature component (*A*) exhibits both polarities and is attributed to chemical remagnetization in Late Mesozoic to Recent times. The high unblocking temperature magnetization (ChRM) of normal and reversed polarity, probably acquired close to the time of deposition of the sediments, produces Cambrian and Ordovician paleomagnetic poles for the NCB and a loosely correlatable magnetic polarity pattern. The paleomagnetic poles appear to pass all seven reliability criteria recently proposed by *Van der Voo* [1989, 1990] and hence are the most reliable Cambrian and Ordovician poles yet obtained for the NCB. Our success in isolating a near-syndepositional magnetization from Early Paleozoic sediments in this study suggests that Paleozoic age rocks of the North China Block may contain other paleomagnetic records useful for refining the APWP of the block and for verifying the low paleolatitudes of major crustal blocks of China during much of the Paleozoic.

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